ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT I

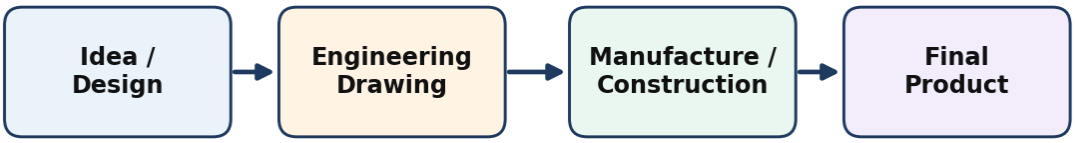
Drawing Basics, Curves & Scales

Learn the alphabet of the language of engineers.

SYLLABUS COVERED

- ▸ Lines, Lettering & Dimensioning
- ▸ Geometrical constructions
- ▸ Regular polygons – general methods
- ▸ Ellipse, Parabola, Hyperbola (general method)
- ▸ Cycloids & Involutes
- ▸ Normal and tangent to curves
- ▸ Plain, Diagonal & Vernier scales

Engineering drawing is the universal language of engineers



Lines • Lettering • Dimensions • Views • Sections

Fig. 0.1 — Why we study Engineering Graphics

1. Introduction & Drawing Essentials

DEFINITION

Engineering Drawing is a graphical language that communicates the **shape, size, material and manufacturing details** of an object accurately and unambiguously, following standard rules (**BIS – SP 46:2003**).

Instrument	Use	Tip
Drawing board + Mini-drafter / T-square	Fix sheet; draw horizontal & inclined lines	Keep head firmly on board edge
Set squares (45° & 30°–60°)	Vertical lines; angles 15°, 30°, 45°, 60°, 75°, 90°	Combine both to get 15° steps
Compass & divider	Draw circles / arcs; transfer & divide lengths	Needle slightly longer than lead
Pencils H, 2H, HB	2H – construction; H – dimension/centre; HB – outlines & lettering	Sharp conical point
French curves, protractor, eraser	Draw smooth curves; measure angles; clean up	Join plotted points freely

Sheet	A0	A1	A2	A3	A4
Size (mm)	841 × 1189	594 × 841	420 × 594	297 × 420	210 × 297

Sheet rule: each next size is obtained by folding the previous size in half (A_{n+1} = ½ A_n); aspect ratio 1 : √2. Every sheet has a **border**, a **title block** (bottom-right) and a **first/third-angle symbol**.

2. Types of Lines (BIS)

Type	Sample line	Name	Application
A	———	Continuous thick	Visible outlines, visible edges
B	———	Continuous thin	Dimension, extension, leader, projection lines; hatching
C	~~~	Continuous thin, freehand	Short break lines
D	—/\/—	Continuous thin with zig-zag	Long break lines
E	- - - - - -	Dashed (thin / medium)	Hidden outlines & hidden edges
F	—·—·—·—	Chain thin (long-dash dot)	Centre lines, lines of symmetry, locus
G	▬·—·—·▬	Chain thin, thick at ends	Cutting-plane lines
H	—··—··—	Chain thin double-dash	Adjacent parts, alternate positions

Fig. 2.1 — Types of lines and where they are used

KEY POINTS

- **Priority** when lines coincide: visible outline → hidden line → cutting plane → centre line → dimension line.
- Thick : thin line ratio ≈ **2 : 1** (e.g. 0.7 mm : 0.35 mm).
- Centre lines extend **~3 mm** beyond the outline and end with a long dash; hidden lines meet at corners with dashes.

3. Lettering

Lettering gives **names, notes and figures** on a drawing. It must be **uniform, legible, single-stroke** and fast to write. Two styles exist: **vertical** and **inclined (75°)**; all capitals are recommended.

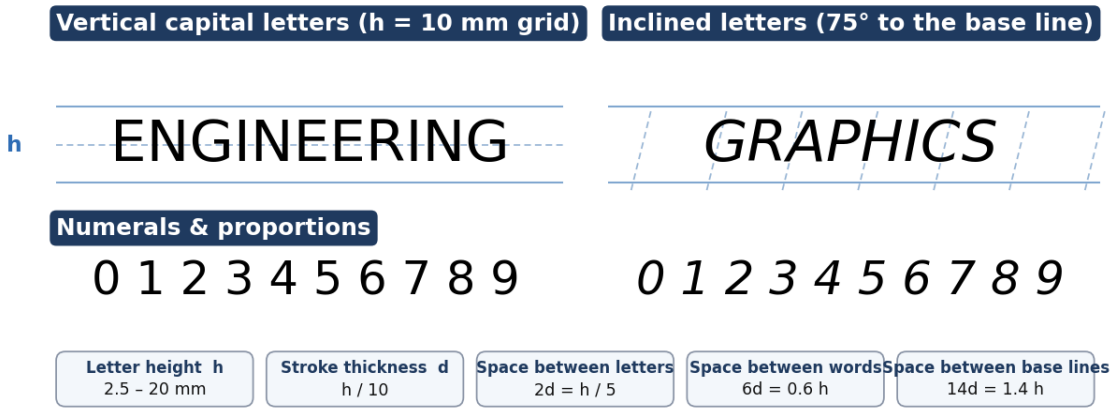


Fig. 3.1 — Single-stroke lettering and standard proportions (Type B: h = 10 d)

Use on drawing	Typical letter height (h)
Main titles / drawing number	10 – 14 mm
Sub-titles, headings	5 – 7 mm
Notes, dimension figures, tolerances	2.5 – 3.5 mm

4. Dimensioning

DEFINITION

Dimensioning is the process of giving **size and location information** on a drawing so that the part can be manufactured without any further questions.

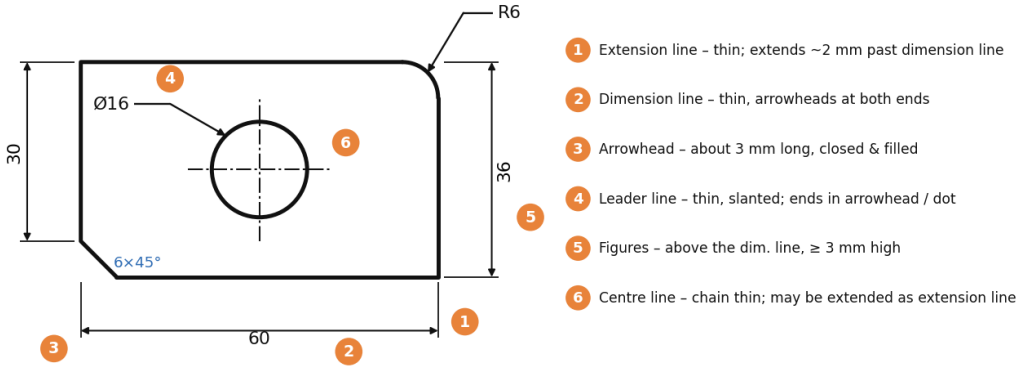


Fig. 4.1 — Elements of dimensioning

Golden rules

- Place dimensions **outside** the view; avoid crossing dimension lines; keep the **smaller dimension nearest** the outline.
- Give each dimension **only once**; never use a centre line, outline or extension line as a dimension line.
- Use symbols: **Ø** diameter, **R** radius, □ square, **SØ / SR** sphere; all in **mm** without writing the unit.
- Extension lines are drawn ⊥ to the feature, ~**2 mm beyond** the dimension line, with a ~1 mm gap from the outline.

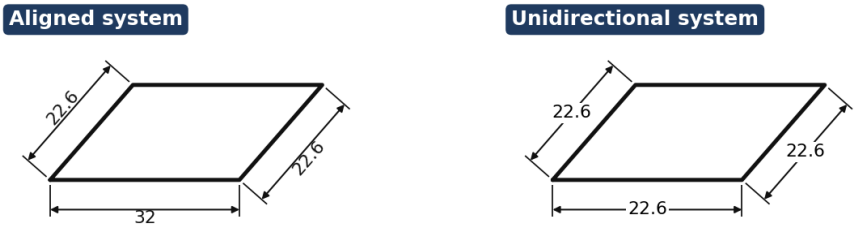
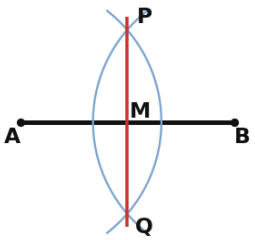


Fig. 4.2 — Aligned and unidirectional systems of placing figures

System	Rule for figures	Where used
Aligned	Parallel to dim. line, readable from bottom or right	Most general drawings
Unidirectional	All figures horizontal; dim. line broken at middle	Large drawings, aircraft/automobile

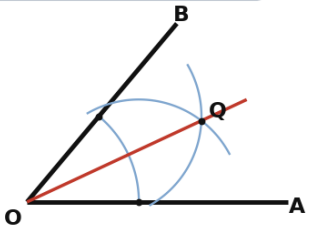
5. Geometrical Constructions

(a) Bisect a line AB



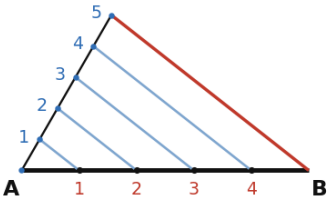
AM = MB, PQ ⊥ AB

(b) Bisect angle AOB



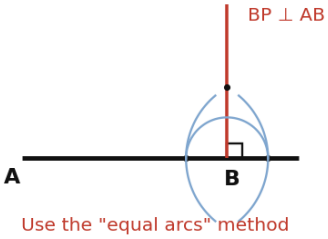
∠AOQ = ∠QOB

(c) Divide AB into 5 equal parts



Join 5–B, then draw ∥ lines

(d) Perpendicular at end B of AB



Use the "equal arcs" method

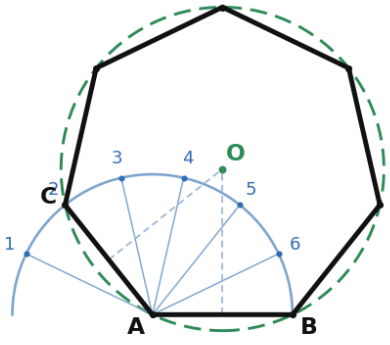
Fig. 5.1 — Basic geometrical constructions

Angle by set squares	Combination
30°, 60°, 90°	30°–60° set square
45°	45° set square
15° / 75°	45° and 30°–60° together (45° − 30° / 45° + 30°)
105° / 120°	45° + 60° / 60° + 60°

6. Regular Polygons – General Methods

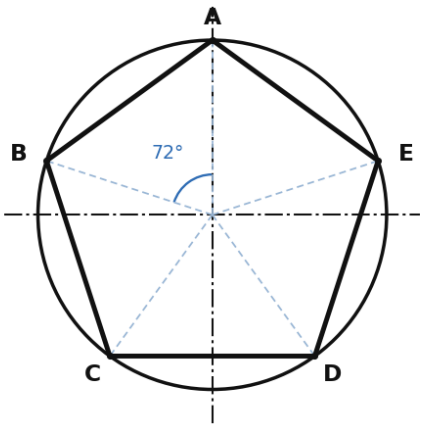
A **regular polygon** has all sides and all interior angles equal; all its vertices lie on a circle. **General method** = one construction that works for *any* number of sides n.

(a) Heptagon on a given side AB



Semicircle on AB ÷ 7 → join A–2 (from far end)

(b) Polygon inside a circle (n = 5)



chord a = 2R·sin(180°/n)

central angle = 360°/n

Fig. 6.1 — General method for a polygon on a given side, and polygon inside a circle

PROCEDURE

1. Draw the given side **AB**. With centre A and radius AB draw a **semicircle** starting from B.
2. Divide the semicircle into **n equal parts** (n = number of sides) and number the points 1, 2, 3 … from the end **far from B**.
3. Join **A–2**; this is the direction of the second side. Mark **AC = AB** on it.
4. Draw perpendicular bisectors of AB and AC – they meet at the **centre O** of the polygon.
5. With O as centre and OA as radius draw the **circumscribing circle** and step off AB around it. Join the points.

FORMULAS

Interior angle = **(n − 2) × 180° / n** | Exterior angle = **360° / n** | Side a = **2R sin(180°/n)** | Inradius r = **a / (2 tan(180°/n))**

Area = **n a² / (4 tan(180°/n))** | Number of diagonals = **n(n − 3)/2**

Sides n	3	4	5	6	7	8
Name	Triangle	Square	Pentagon	Hexagon	Heptagon	Octagon
Interior angle	60°	90°	108°	120°	128.57°	135°
a / R	1.732	1.414	1.176	1.000	0.868	0.765

7. Engineering Curves – Conic Sections

DEFINITION

A **conic** is the locus of a point P that moves so that the ratio of its distance from a fixed point **(focus F)** to its distance from a fixed straight line **(directrix D)** is constant. This ratio is the **eccentricity** e = PF / PN.

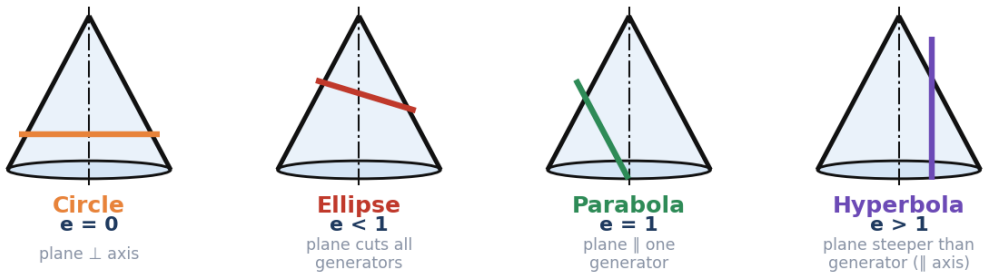


Fig. 7.1 — Conics are plane sections of a cone

Curve	e	Standard equation	Real-life use
Ellipse	< 1	$x^2/a^2 + y^2/b^2 = 1$	Planet orbits, elliptical arches, boiler manholes
Parabola	= 1	$y^2 = 4ax$	Headlight / antenna reflectors, suspension-bridge cables
Hyperbola	> 1	$x^2/a^2 - y^2/b^2 = 1$	Cooling towers, sonic-boom curves, LORAN navigation
Rect. hyperbola	= √2	$x \cdot y = c$	Boyle's law PV = constant

6.1 Ellipse

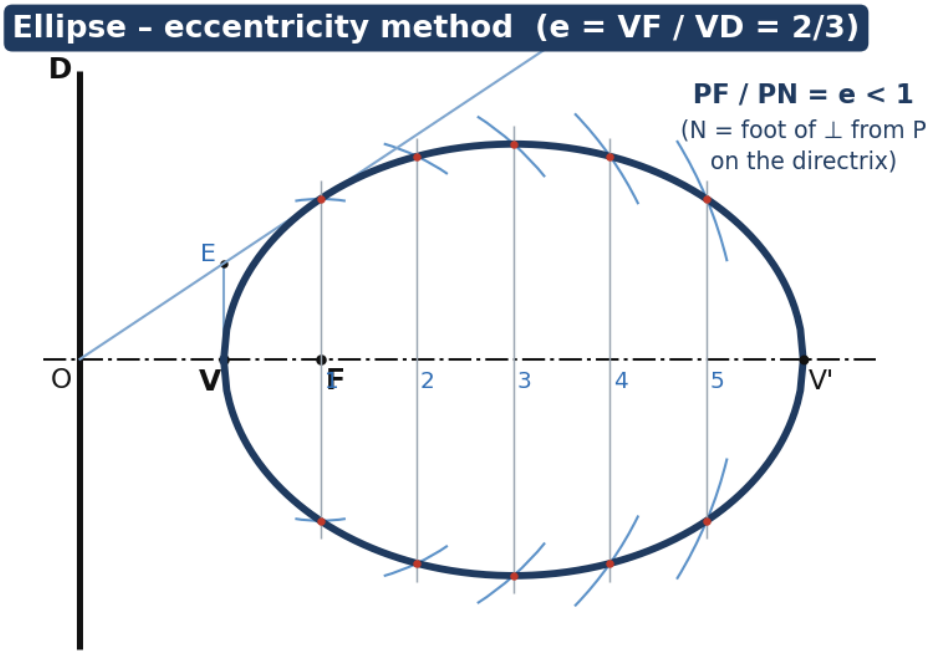


Fig. 7.2 — Ellipse by the general (eccentricity) method

PROCEDURE

1. Draw directrix **DD** and the axis ⊥ to it, meeting at **O**. Mark focus **F** on the axis (e.g. OF = 40 mm).
2. For e = 2/3 divide OF into **2 + 3 = 5** equal parts; vertex **V** is at the 3rd part from O (VF/VO = 2/3).
3. At V erect **VE = VF** and join **O–E** (slope = e). Mark points 1, 2, 3 … on the axis and draw verticals.
4. With **F as centre and radius = (vertical to line OE)** cut the vertical above & below the axis to get points on the curve.
5. Join the points with a **French curve**. The second vertex V′ is found by the same rule on the other side of F.

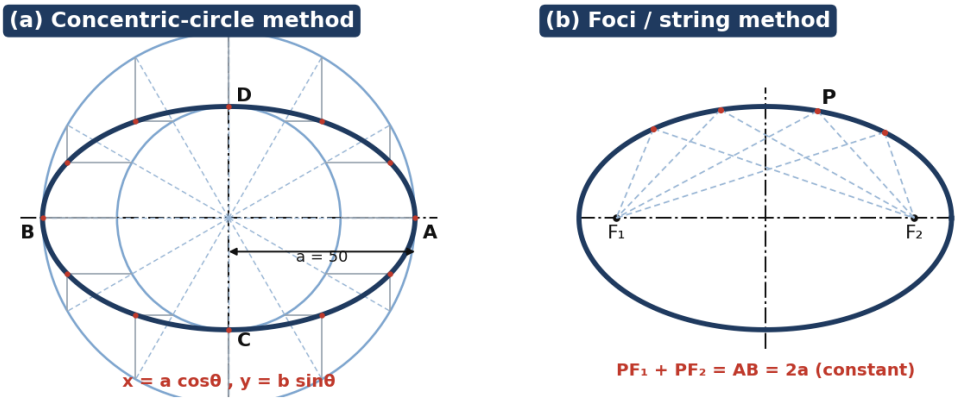


Fig. 7.3 — Other methods: concentric circles & foci (string) method

FORMULAS

Eccentricity **e = √(1 − b²/a²)** | Foci at **±ae** from centre | Sum of focal distances **PF₁ + PF₂ = 2a** | Major axis 2a, minor axis 2b

6.2 Parabola

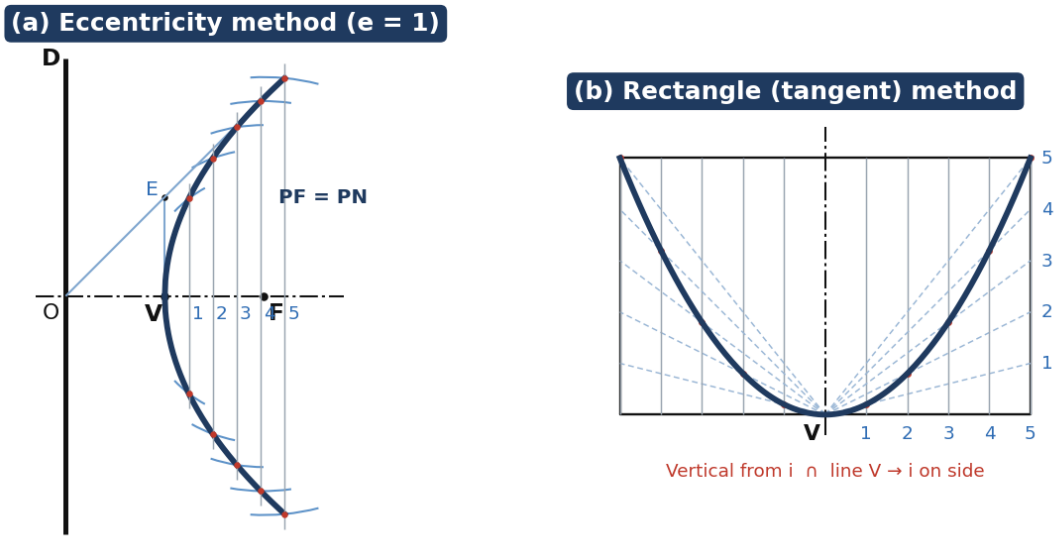


Fig. 7.4 — Parabola – eccentricity method and rectangle method

Eccentricity method: same as the ellipse, but e = 1, so the vertex V is the **mid-point of OF** and VE = VF = VO. Radius for every point = distance of the vertical from the directrix (PF = PN). **Rectangle method:** use when the base and axis lengths are given; divide the half-base and the axis into the same number of equal parts and join the vertex to the division points.

6.3 Hyperbola

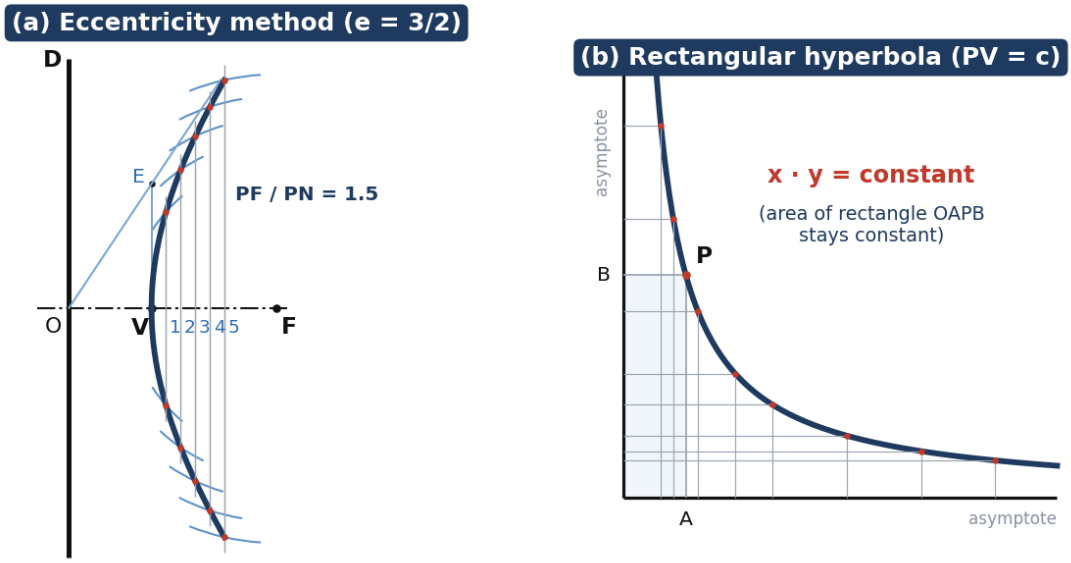


Fig. 7.5 — Hyperbola – eccentricity method and rectangular hyperbola

Eccentricity method: for e = 3/2 divide OF into **2 + 3** parts so that VO : VF = 2 : 3 (VF > VO). Points are found by the same “vertical + arc from F” step and the curve opens away from the directrix. **Rectangular hyperbola:** asymptotes are at 90°; draw rectangle OAPB through P and, through any point on PA/PB lines, the intersection of the vertical and horizontal gives points of the curve (x · y = constant).

8. Cycloidal Curves

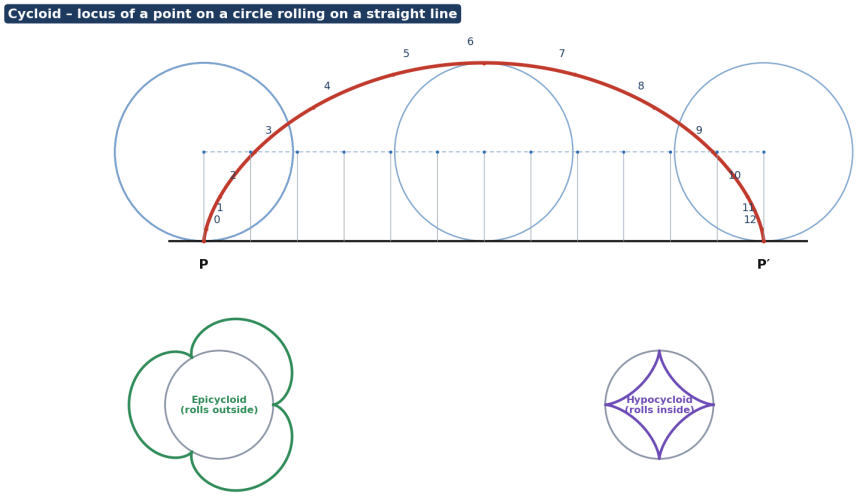


Fig. 8.1 — Cycloid, epicycloid and hypocycloid

Curve	Generating circle rolls …	Directing line / circle
Cycloid	on a straight line (without slipping)	Straight line
Epicycloid	on the **outside** of another circle	Directing circle (radius R)
Hypocycloid	on the **inside** of another circle	Directing circle (radius R)

PROCEDURE

1. Draw the generating circle (radius r) and a base line of length **πD = 2πr**; divide both into **12 equal parts**.
2. Draw the line of centres parallel to the base and locate the successive centres C_1, C_2 … C_{12} by verticals.
3. Through each division of the circle draw a **horizontal line**. With centre C_k and radius r cut horizontal k at $\mathbf{P_k}$.
4. Join P_0, P_1 … P_{12} by a smooth curve.

FORMULAS

Cycloid: **x = r(θ − sin θ), y = r(1 − cos θ)** | Base length of one arch = **2πr** | Height = **2r** | Arc length = **8r**

Epi/Hypocycloid: angle subtended at centre of directing circle **θ = (r / R) × 360°**

9. Involute

DEFINITION

An **involute** is the locus of the free end of a **taut string** unwound from a circle (or polygon). At any point, the **normal to the involute is tangent to the base circle**.

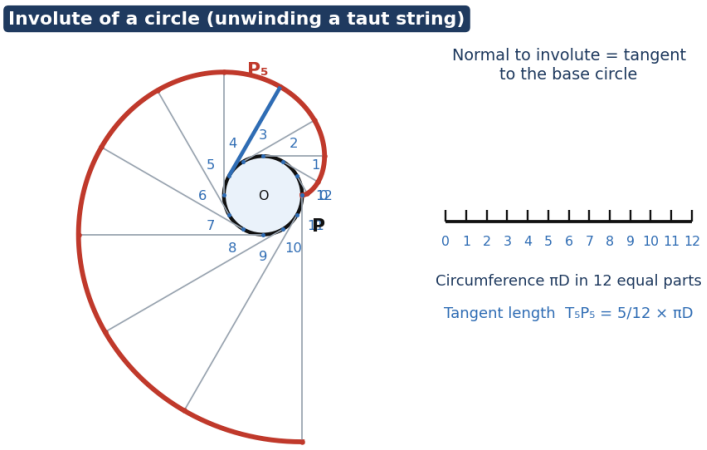


Fig. 9.1 — Involute of a circle

PROCEDURE

1. Draw the base circle (radius r) and divide its circumference into **12 equal parts**; number them 1 … 12.
2. At each point draw the **tangent** (⊥ to the radius) in the same sense.
3. Make tangent at k equal to **k/12 of the circumference** (k × πD/12) – i.e. the length of the arc P–k.
4. Join the end-points by a smooth curve.

APPLICATIONS

Involute profiles are used for **gear teeth** – they give constant velocity ratio and smooth meshing. Cycloids are used in clock/watch gears and pump rotors.

10. Normal & Tangent to Curves

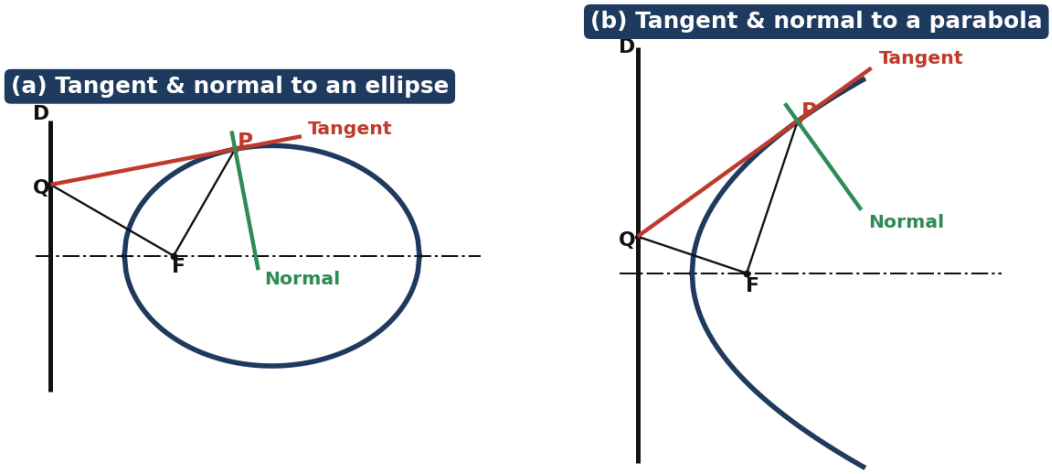


Fig. 10.1 — Tangent and normal using the focus–directrix property

PROCEDURE

Conics (ellipse, parabola, hyperbola) at point P:

1. Join **P to the focus F**.
2. At F draw a line **perpendicular to PF** to meet the directrix at **Q**.
3. Join **QP** and extend – this is the **tangent**.
4. Draw a line **⊥ to the tangent at P** – this is the **normal**.

Curve	Normal at a point P
Circle	Passes through the centre
Involute	Tangent to the base circle
Cycloid	Joins P to the point where the generating circle (through P) touches the base line
Any curve	Draw a circle at P, mark where it cuts the curve on both sides; their chord's ⊥ bisector gives the normal

11. Scales

DEFINITION

A **scale** is a ruler on which actual lengths are shown reduced or enlarged. **Representative Fraction (RF)** = length of drawing / actual length (same units).

FORMULAS

RF = **1 / (ground length per 1 cm of drawing in cm)** | Length of scale = **RF × maximum length to be measured**

Reduced scale RF < 1 (1:50) • **Full** size RF = 1 • **Enlarged** RF > 1 (5:1)

Scale	Reads	Principle
Plain	2 consecutive units (m & dm)	Extra main division at left of 0 divided into sub-divisions
Diagonal	3 units (m, dm, cm)	Similar triangles: diagonals × 10 horizontal lines
Vernier	2–3 units, very fine (least count)	n vernier divisions = (n − 1) main divisions

8.1 Plain scale

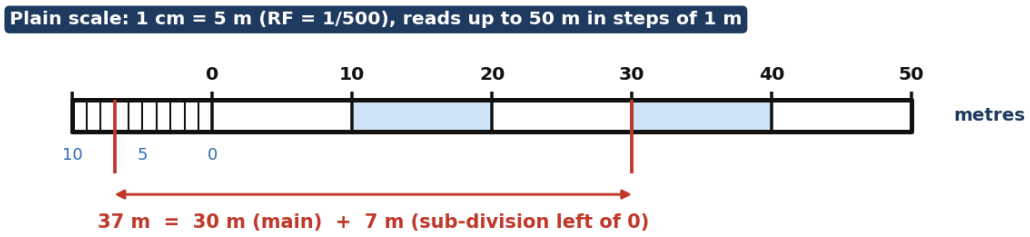


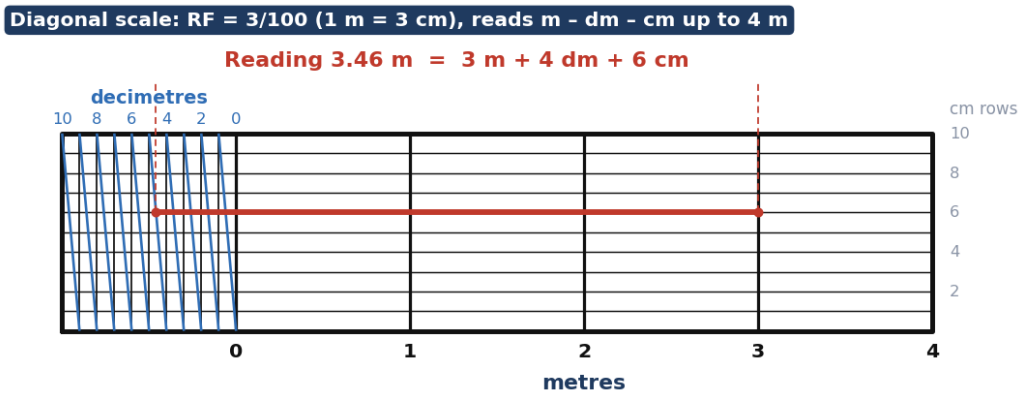
Fig. 11.1 — Plain scale

SOLVED EXAMPLE

Draw a plain scale of 1 cm = 5 m to read up to 50 m; mark 37 m.

RF = 1 cm / 500 cm = **1/500**. Length = 50 m ÷ 5 = **10 cm**. Draw 10 cm, divide into 5 main parts (10 m each = 2 cm). Divide the extra left division into 10 parts (1 m each). Mark 7 m on the left and 30 m on the right: distance = **37 m**.

8.2 Diagonal scale



Each diagonal moves 1 dm-division (3 mm) over 10 rows, so every row adds 1/10 dm = 1 cm.

Fig. 11.2 — Diagonal scale

SOLVED EXAMPLE

Diagonal scale RF = 3/100 to read m, dm, cm up to 4 m; show 3.46 m.

Length = 3/100 × 4 m = **12 cm**. Draw 12 cm, divide into 4 main divisions (3 cm = 1 m). Draw a rectangle of any height, divided into **10** equal rows; subdivide the left division into 10 parts (dm) and draw diagonals. 3.46 m = 3 m (right vertical) + 4 dm (diagonal no. 4) + 6 cm (row no. 6).

8.3 Vernier scale

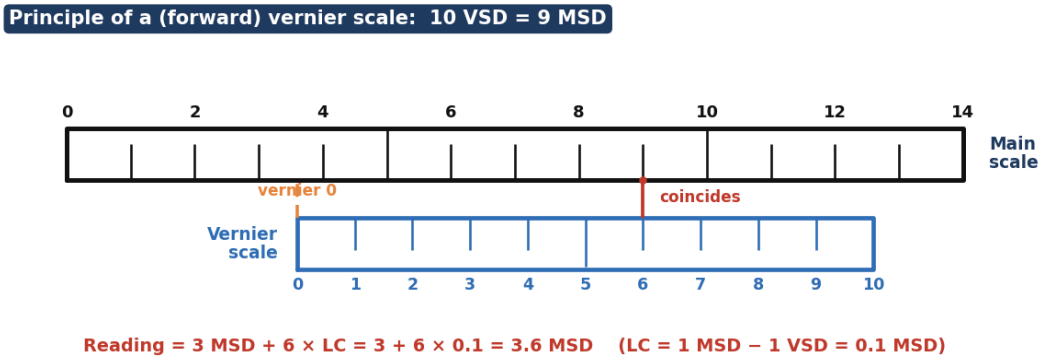


Fig. 11.3 — Vernier scale principle

FORMULAS

Forward vernier: **n VSD = (n − 1) MSD** | Retrograde: **n VSD = (n + 1) MSD** | **Least count LC = 1 MSD − 1 VSD = MSD / n**

Reading = **(main-scale reading) + (coinciding vernier division) × LC**

SOLVED EXAMPLE

Vernier scale RF 1/25 to read 0.01 m on 4 m. 1 m = 4 cm → length = **16 cm**; 1 MSD (0.1 m) = 4 mm; LC = 0.01 m → n = 10; vernier length = 9 MSD = **36 mm** divided into 10 equal parts (3.6 mm each).

12. Quick Revision & Exam Questions

QUICK SUMMARY

- **Lines:** thick – outline; thin – dimension/extension; dashed – hidden; chain – centre/cutting plane.
- **Lettering:** single stroke, h = 10d, inclined at 75°, spacing 2d.
- **Polygons:** semicircle ÷ n → join A–2; interior angle = (n−2)·180/n.
- **Conics:** e<1 ellipse, e=1 parabola, e>1 hyperbola; PF = e·PN; tangent via perpendicular at F.
- **Cycloid** – circle rolling on a line (base = πD); **involute** – string unwound (tangent = arc length).
- **Scales:** RF = drawing/actual; length = RF × max length; LC = MSD/n.

EXAM-ORIENTED QUESTIONS

Q1. List the types of lines used in engineering drawing with one application each. **[2]**

Q2. State the principles of dimensioning and explain aligned and unidirectional systems. **[4]**

Q3. Construct a regular heptagon of side 30 mm by the general method. **[6]**

Q4. Draw an ellipse by the eccentricity method when the distance of focus from directrix is 50 mm and e = 3/4. **[8]**

Q5. Construct a parabola with focus 40 mm from the directrix. Draw tangent and normal at a point 50 mm from the focus. **[8]**

Q6. Draw a rectangular hyperbola when a point P is 20 mm and 40 mm from the two asymptotes. **[8]**

Q7. Draw a cycloid for a circle of 50 mm diameter rolling on a straight line. Draw the normal at a point 40 mm above the base. **[8]**

Q8. Draw an involute of a circle of 40 mm diameter; draw a tangent and normal at a point 100 mm from the centre. **[8]**

Q9. Construct a diagonal scale of RF = 1/40 to read metres, decimetres and centimetres up to 5 m. Show 3.57 m. **[6]**

Q10. Draw a vernier scale of RF 1/20 to read up to 3 m with a least count of 0.01 m. **[6]**

ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT II

Orthographic Projections of Points, Lines & Planes

See a 3-D object as two flat views.

SYLLABUS COVERED

- ▸ Reference planes & quadrants
- ▸ Projections of a point (4 quadrants)
- ▸ Lines ∥ / ⊥ / inclined to reference planes
- ▸ Line inclined to both planes
- ▸ Planes ⊥ both, ∥ one & inclined to other
- ▸ Plane inclined to both reference planes

1. Orthographic Projection – The Basics

DEFINITION

Orthographic projection represents a 3-D object by **two or more 2-D views** obtained by dropping **projectors perpendicular** to the reference planes. The view on the vertical plane is the **Front View (FV)**; on the horizontal plane the **Top View (TV)**.

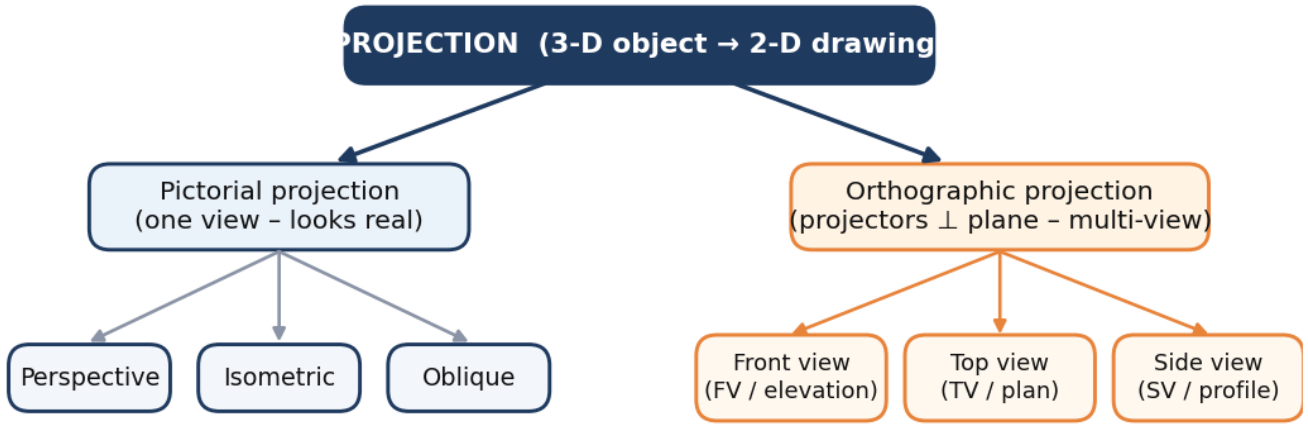


Fig. 1.1 — Classification of projections

Term	Meaning
Plane of projection	Plane on which the view is obtained (HP, VP, PP)
Projector	Imaginary line from the object point ⊥ to the plane
HP / VP / PP	Horizontal Plane / Vertical Plane / Profile (side) Plane
XY line	Line of intersection of HP and VP – the **reference line**
Notation	Object point **A**; FV **a′** (dashed-prime); TV **a**; SV **a″**

EXAM TIP

We follow **First-Angle Projection** (BIS/Indian practice): object lies in the **first quadrant**, between the observer and the plane. In **Third-Angle** projection (USA) the plane lies between observer and object.

2. Reference Planes & Four Quadrants

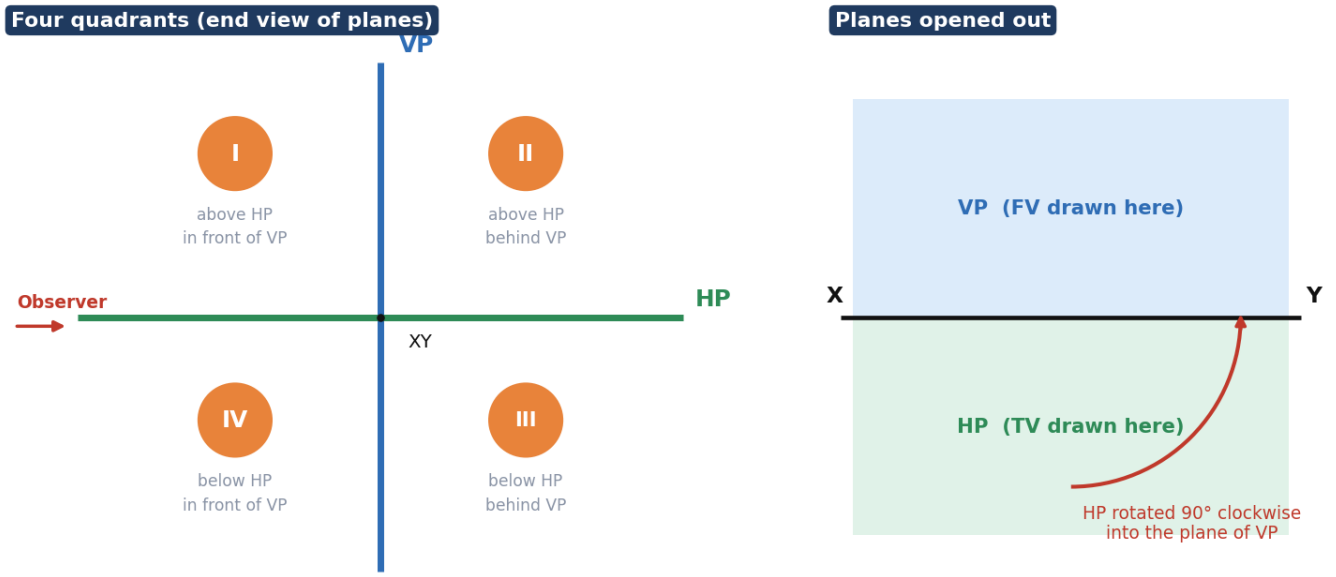


Fig. 2.1 — HP and VP divide space into four quadrants; HP is rotated into the VP to get one flat sheet

- **Importance of XY:** it separates the FV (above XY, on VP) from the TV (below XY, on HP) after rotation; the distance of a view from XY gives the **height** (FV) or **distance in front of VP** (TV).
- The two views of a point always lie on **one vertical projector (⊥ XY)**.
- After rotation: the part of HP **in front** of VP goes **below XY**; the part **behind** VP goes **above XY**.

3. Projection of a Point

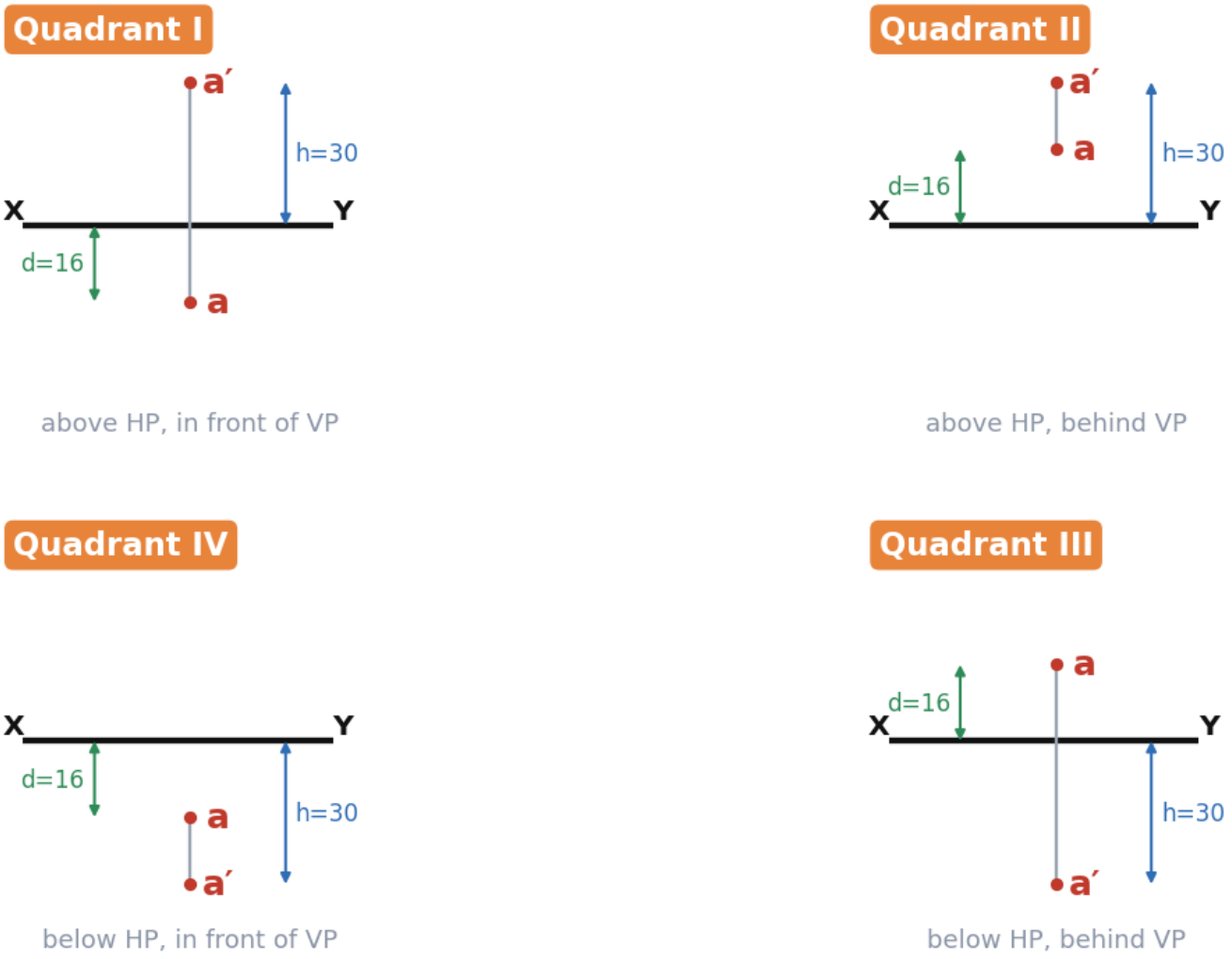


Fig. 3.1 — A point in each quadrant (height above/below HP = 30, distance in front/behind VP = 16)

Quadrant	Position of point	FV a´	TV a
I	above HP, in front of VP	above XY	below XY
II	above HP, behind VP	above XY	**above XY**
III	below HP, behind VP	**below XY**	above XY
IV	below HP, in front of VP	below XY	below XY

KEY POINTS

- FV a´ ⟶ distance from XY = **height of point above/below HP**; TV a ⟶ distance from XY = **distance from VP**.
- Point **in HP**: FV on XY. Point **in VP**: TV on XY. Point in both: both views on XY.
- **Memory trick:** in Quadrants II & IV the two views lie on the *same side* of XY (II – both above, IV – both below).

4. Projections of Straight Lines

A straight line is fixed by the projections of its two end points. Notation: **TL** = true length, **θ** = true inclination to HP, **φ** = true inclination to VP, **α, β** = apparent inclinations in FV and TV.

Position of line	FV	TV
∥ HP and ∥ VP	TL, ∥ XY	TL, ∥ XY
⊥ HP (∥ VP)	TL, ⊥ XY	Point
⊥ VP (∥ HP)	Point	TL, ⊥ XY
Inclined θ to HP, ∥ VP	TL at true θ	Shorter (TL cosθ), ∥ XY
Inclined φ to VP, ∥ HP	Shorter (TL cosφ), ∥ XY	TL at true φ
Inclined to both	Shorter, at α > θ	Shorter, at β > φ

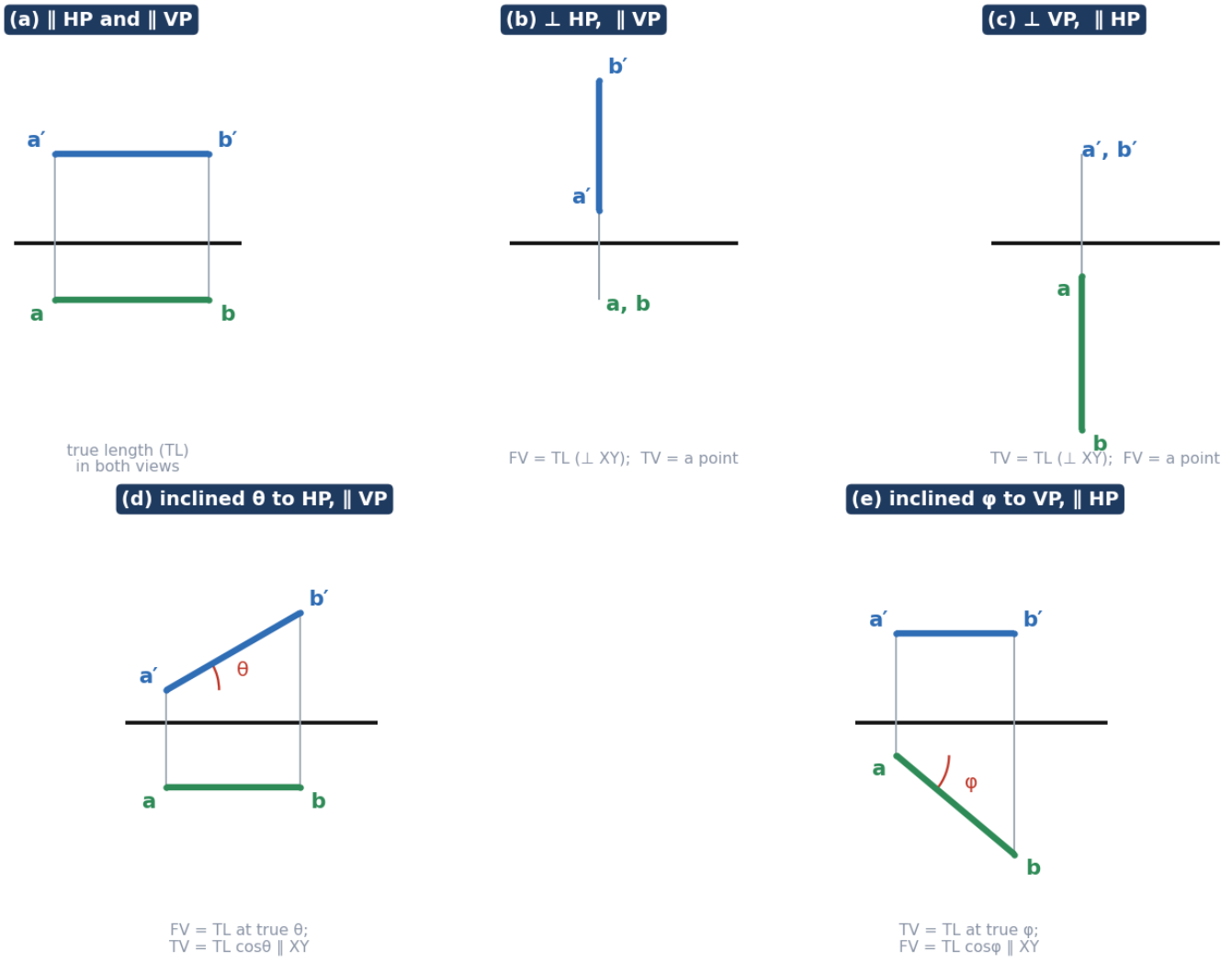


Fig. 4.1 — Lines in special positions

4.1 Line inclined to both reference planes

FORMULAS

TV length = **TL cosθ** | FV length = **TL cosφ** | Height difference of ends = **TL sinθ** | Distance difference = **TL sinφ**

θ + φ ≤ 90° | **α ≥ θ, β ≥ φ** | tanθ = tanα · cosβ | tanφ = tanβ · cosα

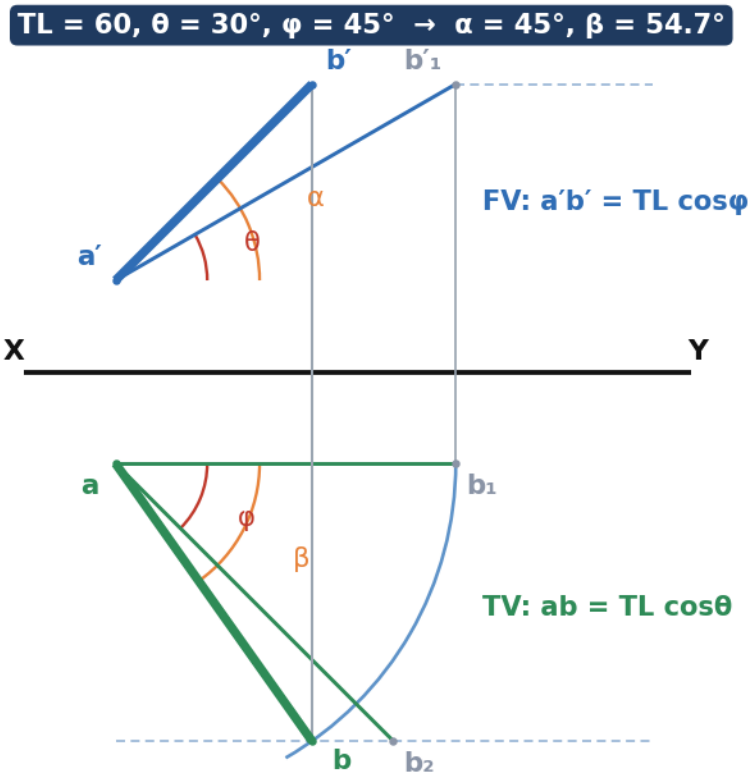


Fig. 4.2 — Line of TL = 60 mm, θ = 30°, φ = 45° (locus method)

PROCEDURE

1. Draw XY. Fix a′ (height) and a (distance). Draw **a′b₁′ = TL at θ**; project TV length **ab₁ = TL cosθ** (horizontal).
2. From a draw **ab₂ = TL at φ** in the TV; the horizontal through b₂ is the **locus of b** (distance = TL sinφ).
3. With centre a and radius **ab₁** draw an arc to cut that locus at **b**.
4. Project b up to the horizontal through b₁′ (locus of b′) to get **b′**. Join a′b′ and ab.

SOLVED EXAMPLE

A line AB, 60 mm long, is inclined 30° to HP and 45° to VP. Find the apparent lengths and angles.

FV = 60 cos45° = **42.4 mm**; TV = 60 cos30° = **52 mm**; height difference = 60 sin30° = **30 mm**; distance difference = 60 sin45° = **42.4 mm**. From the figure: α = **45°**, β = **54.7°**. Check: tanθ = tan45° · cos54.7° = 0.577 ✓.

APPLICATIONS

Line projections are used for **pipelines, roof rafters, struts and trusses** where members lie obliquely in space. **Traces**: the point where a line (extended) meets HP is its **HT** (FV on XY), and meets VP is its **VT** (TV on XY).

5. Projections of Planes

A **plane** is a 2-D figure (triangle, square, pentagon, hexagon, circle…) with negligible thickness. Its position is described by the inclination of its **surface** to the reference planes. Always start with the plane **parallel to a reference plane** (true shape visible) and then tilt it in **stages**.

Position of plane	FV	TV
∥ HP (⊥ VP)	Line ∥ XY	**True shape**
∥ VP (⊥ HP)	**True shape**	Line ∥ XY
⊥ HP and ⊥ VP	Line ⊥ XY	Line ⊥ XY
Inclined θ to HP, ⊥ VP	Line at true θ	Reduced shape
Inclined φ to VP, ⊥ HP	Reduced shape	Line at true φ
Inclined to both	Reduced shape	Reduced shape

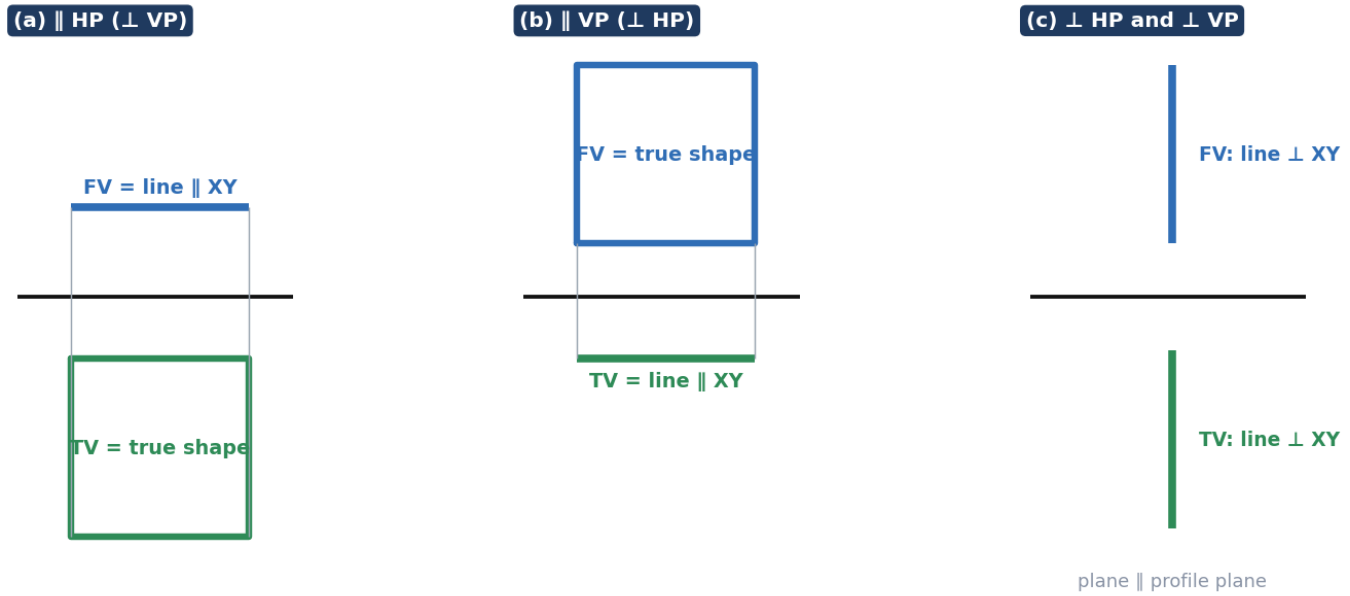


Fig. 5.1 — Planes parallel to a reference plane and perpendicular to both

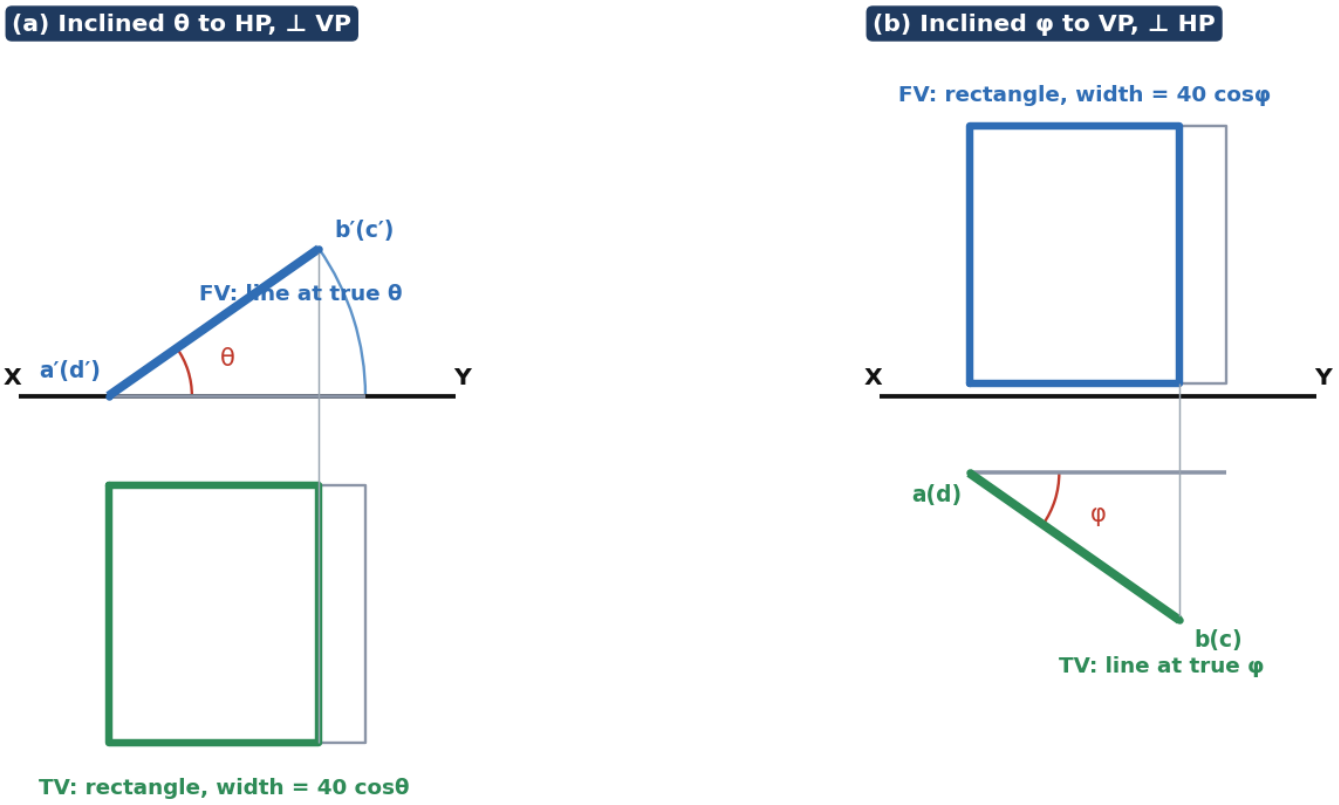


Fig. 5.2 — Plane inclined to one reference plane and perpendicular to the other

FORMULAS

Reduced width = **true width × cos(inclination)** | Area of TV = **true area × cosθ** (plane at θ to HP) | Area of FV = **true area × cosφ**

5.1 Plane inclined to both reference planes

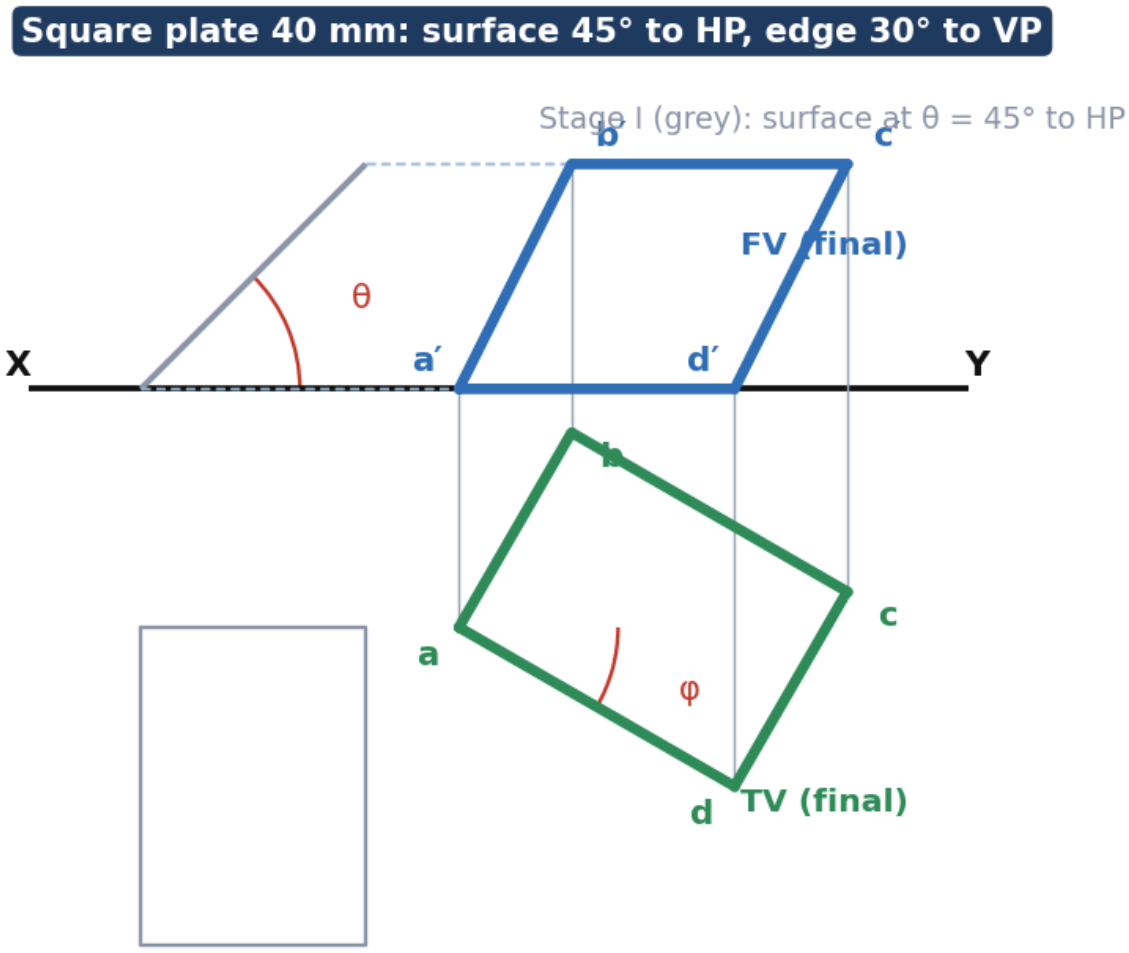


Fig. 5.3 — Two-stage construction for a square plate

PROCEDURE

1. **Stage I:** assume the plane ∥ HP. Draw TV = true shape (one edge ⊥ XY or ∥ XY as given) and FV = line on XY.
2. Tilt the FV about the edge/corner on HP so that it makes **θ with XY**; project the new TV (reduced shape).
3. **Stage II:** redraw the Stage-I TV rotated so that the given edge (or diagonal) makes **φ with XY**. Keep its shape unchanged.
4. Project the final FV: **verticals from the new TV** and **horizontals from the Stage-I FV heights**. Join in order.

EXAM TIP

Golden rule: in a change of position, the **view in which the inclination is shown true is moved rigidly**, and the **other view keeps its heights (or distances)** – they only slide along parallel lines. Hidden edges are drawn as short dashes.

6. Quick Revision & Exam Questions

QUICK SUMMARY

- FV is above XY in Q I & II; TV is below XY in Q I & IV.
- Line ∥ plane → TL in that view; line ⊥ plane → point view.
- TV = TL cosθ, FV = TL cosφ; θ + φ ≤ 90°.
- Plane ∥ plane → true shape; plane ⊥ plane → line view.
- Two-stage method: change **θ first** (FV), then **φ** (TV).

EXAM-ORIENTED QUESTIONS

Q1. Define reference planes and quadrants. Why is the XY line important? **[2]**

Q2. A point P is 30 mm above HP and 25 mm behind VP. Draw its projections and name the quadrant. **[4]**

Q3. Point A is 20 mm below HP and 30 mm in front of VP. Point B is 40 mm above HP and 10 mm behind VP; distance between projectors 50 mm. Draw projections of AB. **[6]**

Q4. A line AB 70 mm long is inclined 30° to HP and 45° to VP. A is 15 mm above HP and 20 mm in front of VP. Draw FV and TV; find α and β. **[8]**

Q5. A 60 mm line is parallel to VP, inclined 40° to HP, end A 10 mm above HP. Draw its projections. **[6]**

Q6. A regular pentagon of side 30 mm has one side on HP; its surface is inclined 45° to HP and ⊥ VP. Draw the projections. **[8]**

Q7. A square plate of 40 mm side has its surface inclined 30° to HP and a diagonal inclined 45° to VP. Draw FV and TV. **[8]**

Q8. Name the position of a plane whose FV and TV are both lines ⊥ XY. **[4]**

ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT III

Projections of Solids

Turn real 3-D objects into FV and TV.

SYLLABUS COVERED

- ▸ Polyhedra & solids of revolution
- ▸ Simple positions of the axis
- ▸ Axis ⊥ HP, ⊥ VP, ∥ both planes
- ▸ Axis inclined to HP, ∥ VP
- ▸ Axis inclined to VP, ∥ HP

1. Types of Solids

DEFINITION

A **solid** has three dimensions – length, breadth and height. In engineering graphics it is represented by its **FV and TV** (and sometimes SV).

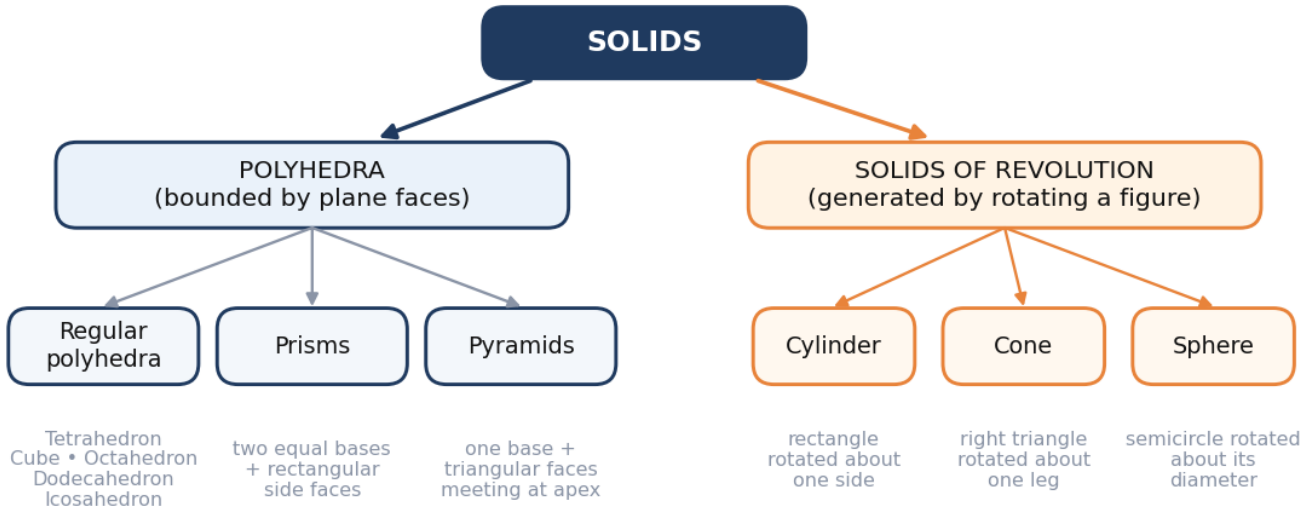
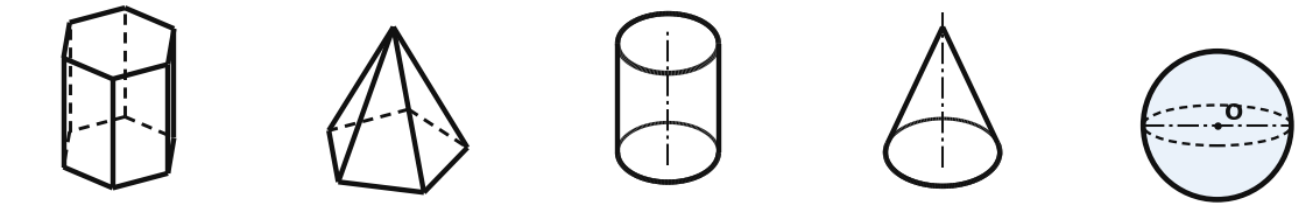


Fig. 1.1 — Classification of solids



Hexagonal prism
2 bases • lateral faces
edges • axis

Pentagonal pyramid
base • apex • slant edges
base edges • axis

Cylinder
2 circular ends
generators • axis

Cone
circular base • apex
generators • axis

Sphere
centre • radius
(no edges/faces)

Fig. 1.2 — Common solids and their parts

Term	Meaning
Base / top	Plane face(s) on which the solid stands / opposite end
Axis	Line joining the centres of the two bases (prism, cylinder) or base-centre to apex (pyramid, cone)
Apex	Point where all slant edges / generators meet
Slant edge (pyramid) / generator (cone)	Edge / line from the apex to a base corner / rim point
Lateral face	Side faces: rectangles (prism), triangles (pyramid)
Right solid	Axis is ⊥ to the base

Regular polyhedron	Faces F	Vertices V	Edges E	Face shape
Tetrahedron	4	4	6	Equilateral triangle
Cube (hexahedron)	6	8	12	Square
Octahedron	8	6	12	Equilateral triangle
Dodecahedron	12	20	30	Regular pentagon
Icosahedron	20	12	30	Equilateral triangle

FORMULAS

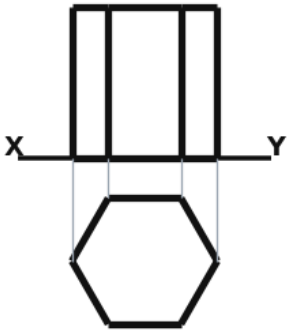
Euler's formula: $F + V - E = 2$ | Prism of n sides: $F = n + 2$, $V = 2n$, $E = 3n$ | Pyramid of n sides: $F = n + 1$, $V = n + 1$, $E = 2n$

2. Projection of Solids in Simple Positions

Simple position = the axis is perpendicular to one reference plane (hence parallel to the other), or parallel to both. The base is then parallel to a reference plane and shows its **true shape**.

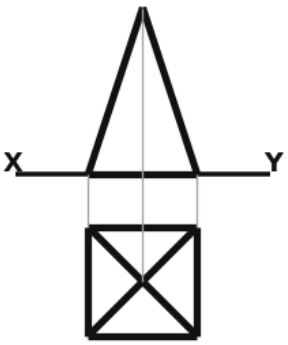
Axis position	Prism / cylinder	Pyramid / cone
⊥ HP	FV rectangle • TV true base	FV triangle • TV true base (+ lines to apex)
⊥ VP	FV true base • TV rectangle	FV true base (+ lines to apex) • TV triangle
∥ HP and ∥ VP	FV rectangle • TV rectangle	FV triangle • TV triangle

(a) Hexagonal prism, axis ⊥ HP



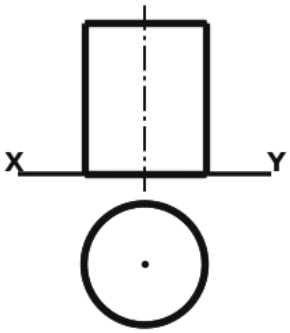
FV: rectangle (3 faces seen)
TV: regular hexagon

(b) Square pyramid, axis ⊥ HP



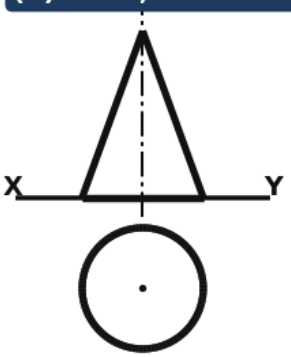
FV: triangle
TV: square + diagonals to apex

(c) Cylinder, axis ⊥ HP



FV: rectangle
TV: circle

(d) Cone, axis ⊥ HP



FV: triangle
TV: circle + centre point

Fig. 2.1 — Axis perpendicular to HP

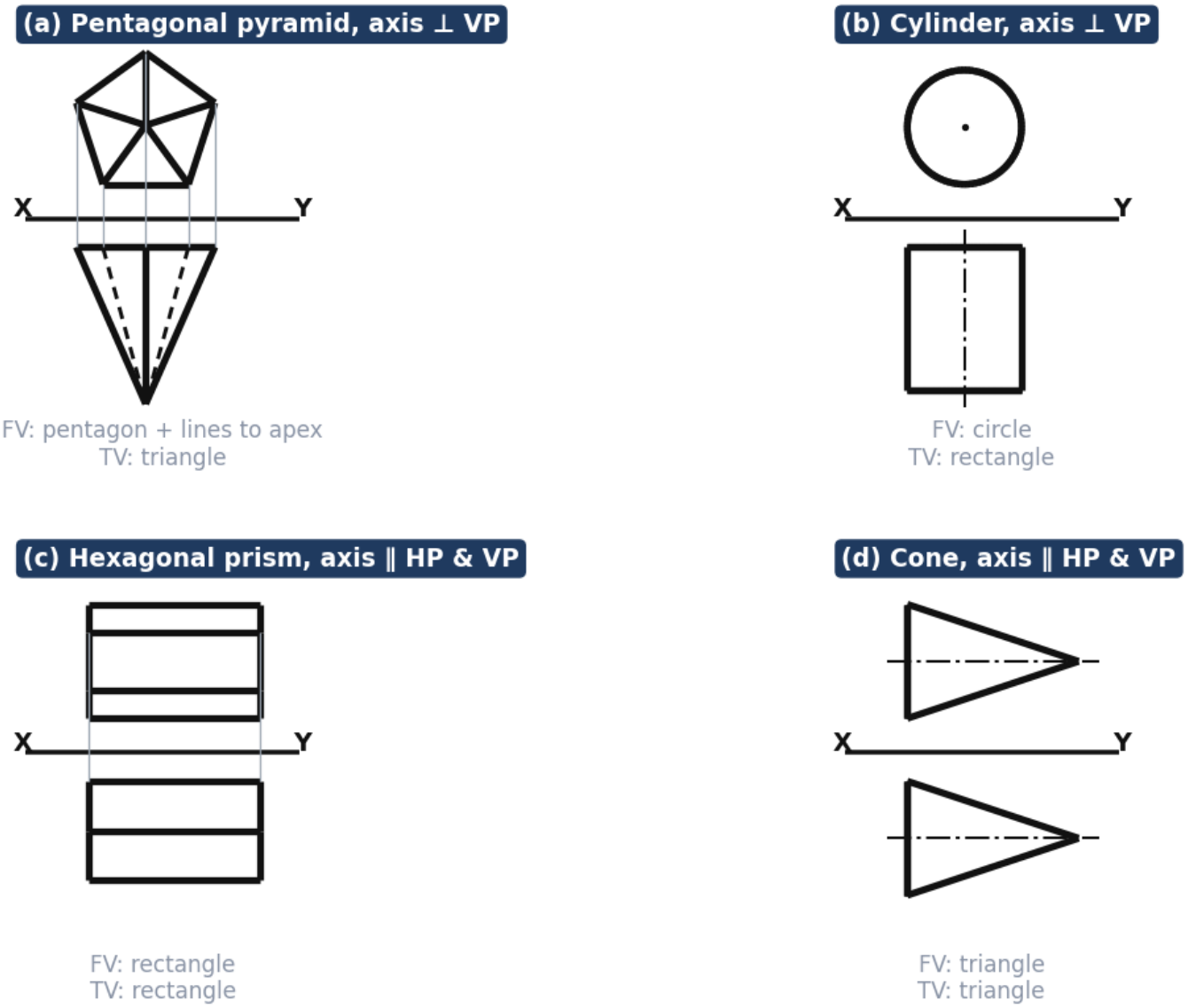


Fig. 2.2 — Axis perpendicular to VP, and axis parallel to both planes

PROCEDURE

1. Draw XY. Mark the **base shape true** in the view where the axis appears as a point.
2. Project the other view using vertical projectors; height of the solid = **axis length**.
3. For a pyramid/cone join the **apex** (centre of base) to all base corners / tangents.
4. Mark **hidden edges dashed** – an edge is hidden when it lies behind a visible face in that view.

3. Axis Inclined to HP and Parallel to VP

Use the **two-stage method**. Stage I: axis ⊥ HP (simple position). Stage II: tilt the **FV** so the axis makes the given angle **θ** with XY, keeping its shape unchanged; the TV is then projected by **verticals from the new FV** and **horizontals from the Stage-I TV**.

Hexagonal prism (side 20, axis 56): base edge on HP

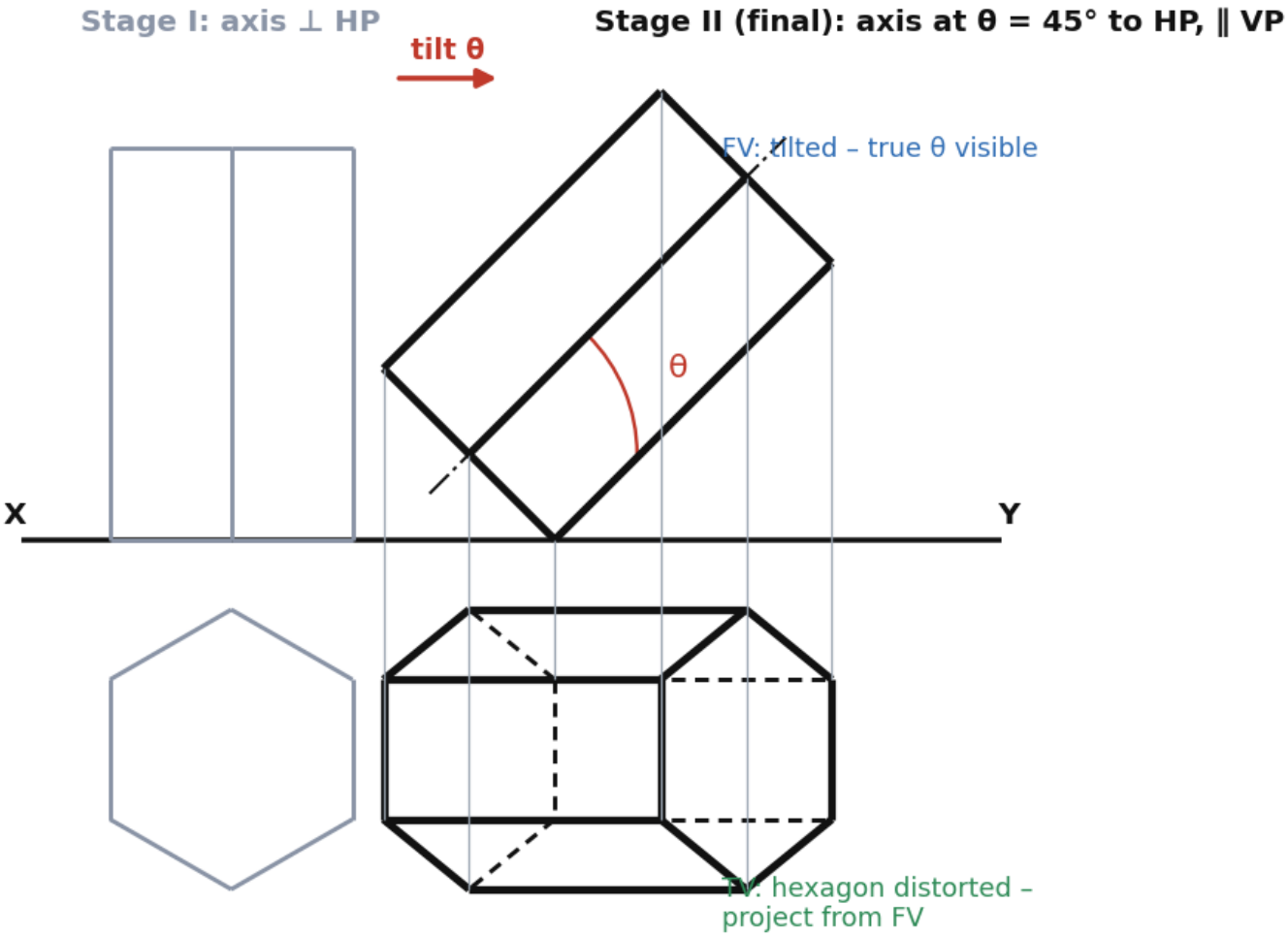


Fig. 3.1 — Hexagonal prism with axis inclined 45° to HP

KEY POINTS

- **Base edge on HP** → in Stage I place that edge ⊥ XY (it appears as a point in FV) at the corner that will touch HP.
- **Base corner on HP** → place a corner at the lowest point of the FV base.
- During tilting, **the width in TV** stays the same; only the **x-spacing** (length along XY) changes.

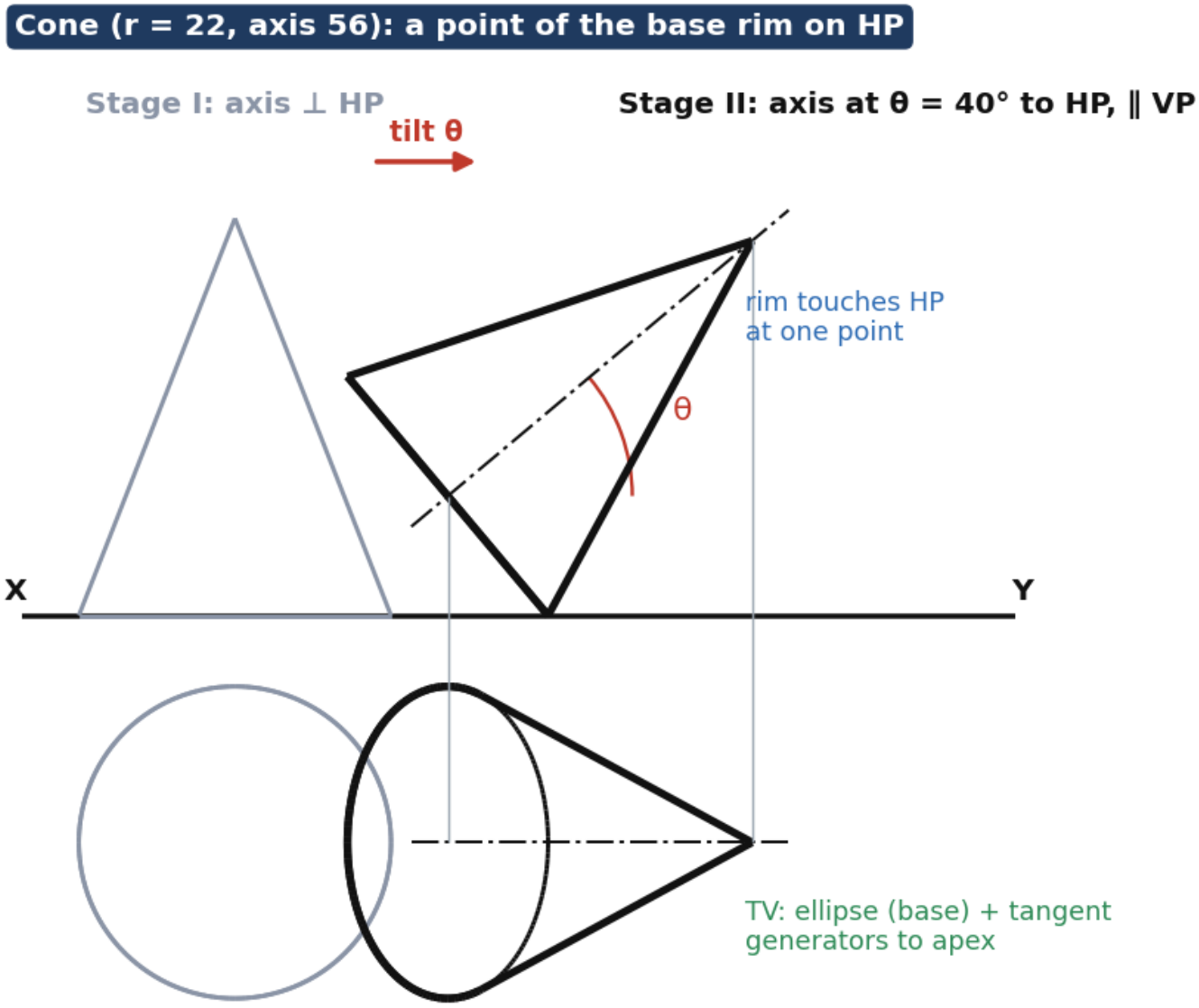


Fig. 3.2 — Cone with a point of the base rim on HP and axis inclined 40° to HP

APPLICATIONS

Typical cases: prism/cylinder resting on an edge of its base, cone resting on a point of its base circle, pyramid resting on a base corner.

4. Axis Inclined to VP and Parallel to HP

Stage I: axis ⊥ **VP** (FV = true base). Stage II: rotate the **TV** so the axis makes the given angle **φ** with XY, then project the FV with **verticals from the new TV** and **horizontals from the Stage-I FV**.

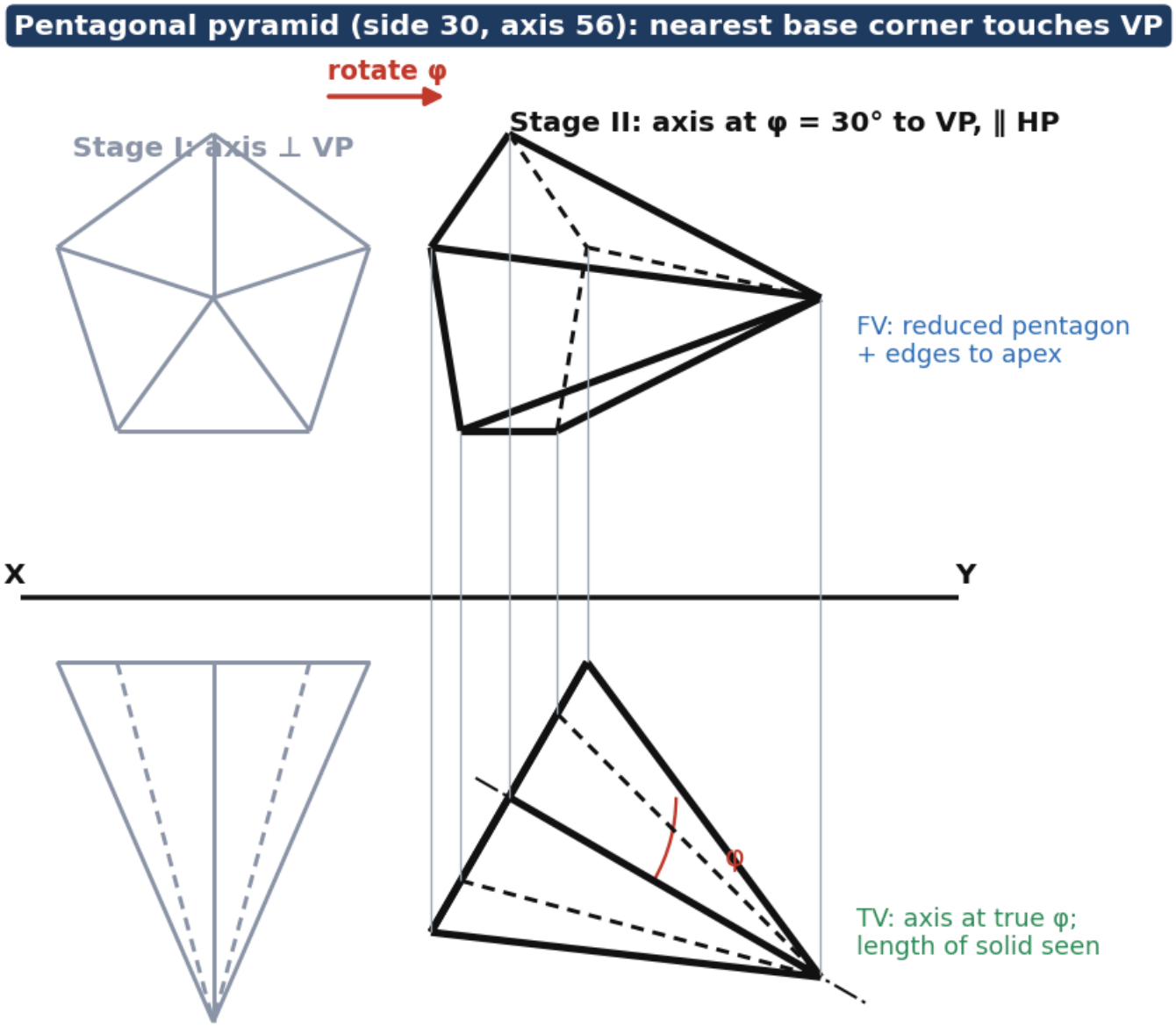


Fig. 4.1 — Pentagonal pyramid with axis inclined 30° to VP

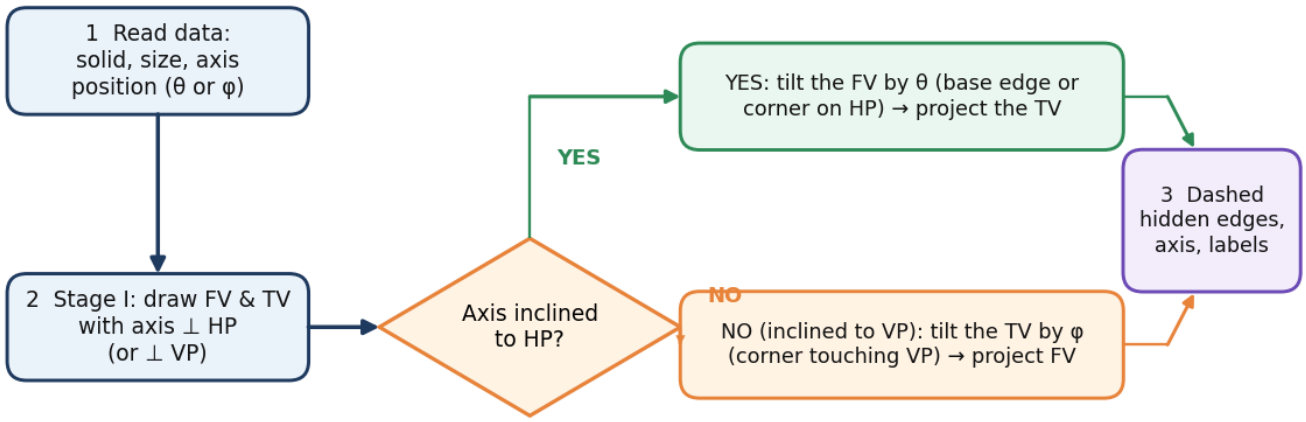


Fig. 4.2 — Procedure to solve problems on solids

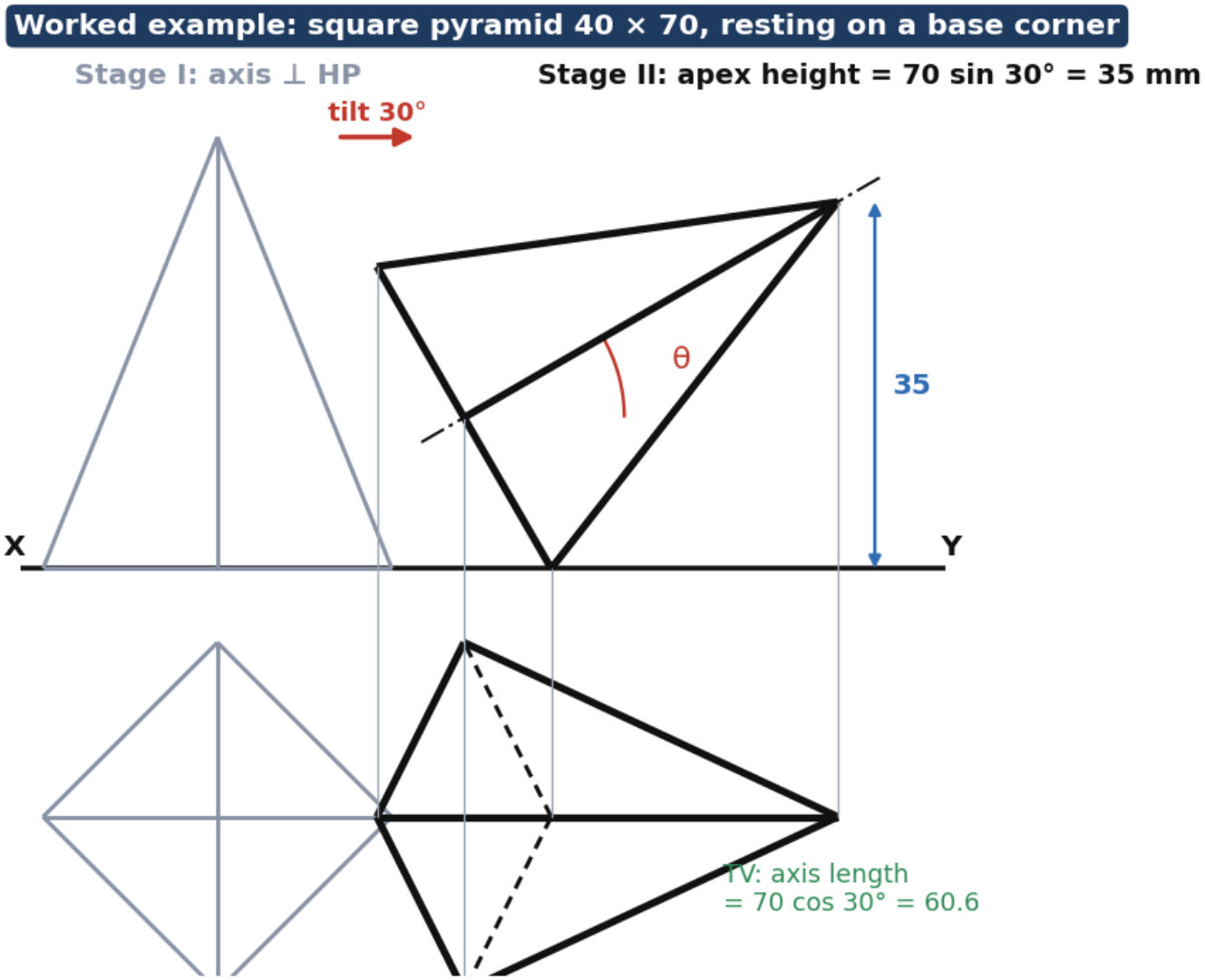


Fig. 4.3 — Worked example (computed to scale)

SOLVED EXAMPLE

A square pyramid, base side 40 mm, axis 70 mm, rests on a base corner so that the axis is inclined 30° to HP (‖ VP). Find the apparent length of the axis.

Stage I: TV square 40 mm, apex at centre; FV triangle base 40 mm, height 70 mm. Stage II: tilt FV by **30°** to XY about the corner on HP. Height of apex = 70 sin30° = **35 mm** above HP; horizontal span of axis in TV = 70 cos30° = **60.6 mm**. The FV shows the **true axis length 70 mm** (axis ‖ VP).

5. Special Cases & Visibility Rules

Situation	What to draw
Sphere (any position)	FV and TV are both circles of the sphere’s diameter; draw centre lines
Cylinder / cone, axis ‖ both planes	Rectangle / triangle in both views; draw the base circle as a line (and a centre line)
Cone with axis inclined	Base circle becomes an **ellipse** in the other view; apex joined by tangents to the ellipse
Frustum / truncated solid	Draw the full solid, then cut parallel to the base; top face has the same shape (scaled)
Solid resting on a face / edge / corner	Lowest feature lies on XY in the view where height is seen

KEY POINTS

- An **edge is visible** if at least one adjacent face faces the observer; otherwise it is **hidden** and drawn dashed.
- In the **FV** the observer looks from the front; in the **TV** from above – so the edge nearest to the observer in that direction is visible.
- Where a visible and a hidden edge coincide, draw the **visible (thick)** line.

6. Formulae & Common Mistakes

Solid	Volume	Total surface area
Prism (n sides, side a, height h)	Base area × h	2 × base area + n a h
Pyramid	⅓ × base area × h	Base area + ½ × perimeter × slant height
Cylinder	π r² h	2π r (r + h)
Cone	⅓ π r² h	π r (r + l), l = √(r² + h²)
Sphere	4⁄3 π r³	4 π r²

EXAM TIP

- Angle **θ** to HP is measured in the **FV**; angle **φ** to VP is measured in the **TV** – never mix them.
- Between stages the **shape does not change**; only the orientation does.
- Do not forget **hidden (dashed) lines** and the **axis (chain thin line)** in the final views.
- If a solid rests on a **point** of its base circle, draw the cone/cylinder so the lowest rim point lies on XY.

QUICK SUMMARY

- Prism/cylinder: FV & TV rectangles in some position; pyramid/cone: triangles.
- Axis ⊥ plane → that view is the true base.
- Inclined to HP → tilt **FV**; inclined to VP → tilt **TV**.
- Always: Stage I (simple) → Stage II (tilt) → project → hidden lines.

EXAM-ORIENTED QUESTIONS

Q1. Classify solids with examples. State Euler's formula and verify it for a hexagonal prism. **[4]**

Q2. Draw the projections of a pentagonal prism of base side 30 mm and axis 60 mm, axis ⊥ HP, with a base edge ∥ VP. **[6]**

Q3. Draw the projections of a cone (base Ø50, axis 70 mm) with axis ∥ HP and ∥ VP. **[6]**

Q4. A hexagonal prism (side 25, height 60) rests on a base edge on HP; its axis is inclined 45° to HP and ∥ VP. Draw FV and TV. **[8]**

Q5. A cone of base diameter 50 mm and axis 70 mm has a point of its base on HP; axis inclined 40° to HP and ∥ VP. Draw the views. **[8]**

Q6. A square pyramid (side 40, axis 65) has its axis inclined 30° to VP and ∥ HP; a corner of the base touches VP. Draw FV and TV. **[8]**

Q7. A cylinder Ø40 × 60 lies on HP with its axis inclined 30° to VP. Draw its projections. **[6]**

ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT IV

Sections of Solids & Development of Surfaces

Cut it open, then unfold it flat.

SYLLABUS COVERED

- ▸ Perpendicular & inclined section planes
- ▸ Sectional views & true shape of section
- ▸ Sections of solids in simple positions
- ▸ Parallel-line & radial-line development
- ▸ Development: cube, prism, cylinder, pyramid, cone

1. Sections of Solids – Basics

DEFINITION

Section plane (cutting plane) – an imaginary plane that cuts the solid. **Section** – the figure formed on the cut surface. **Sectional view** – the view of the remaining solid after the portion on the observer's side is removed. **True shape of section** – the actual shape of the cut surface, seen when looking ⊥ to the cutting plane.

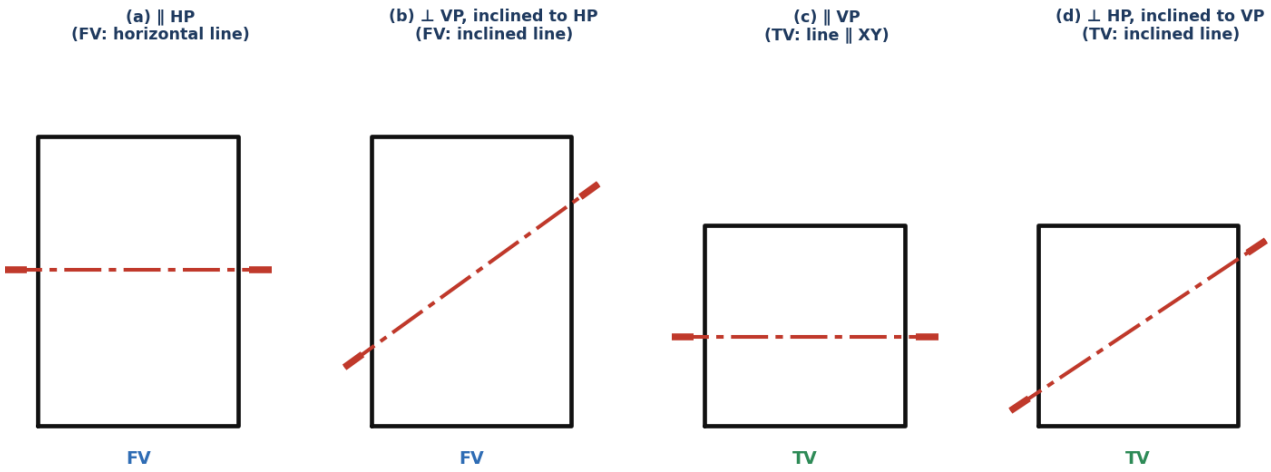


Fig. 1.1 — Common positions of cutting planes (red chain line = trace of the cutting plane)

Section plane	Seen as a line in	True shape of section seen in
‖ HP (⊥ VP)	FV (horizontal line)	**TV** – directly
‖ VP (⊥ HP)	TV (line ‖ XY)	**FV** – directly
⊥ VP, inclined θ to HP (AIP)	FV (inclined line)	Auxiliary view on a plane ‖ cutting plane
⊥ HP, inclined φ to VP (AVP)	TV (inclined line)	Auxiliary view on a plane ‖ cutting plane

PROCEDURE

1. Draw FV and TV of the **solid in simple position**. Draw the **cutting-plane trace** in the view where it appears as a line (chain line, thick at ends).
2. Mark where the trace meets each **edge / generator** in that view: points **1, 2, 3 …** Name them on the trace.
3. Project these points to the other view (verticals / horizontals) on the corresponding edges to get the **section points**; join in order.
4. **Remove** the portion between the observer and the plane; draw the remaining outline. **Hatch** the section at 45°.
5. For the **true shape**: draw X_1Y_1 ‖ trace, project ⊥ to the trace from each point and make distances = those in the other view from XY.

2. Sections of Prism, Pyramid, Cylinder & Cone

Square prism (40 × 70) cut by a plane ⊥ VP inclined 40° to HP

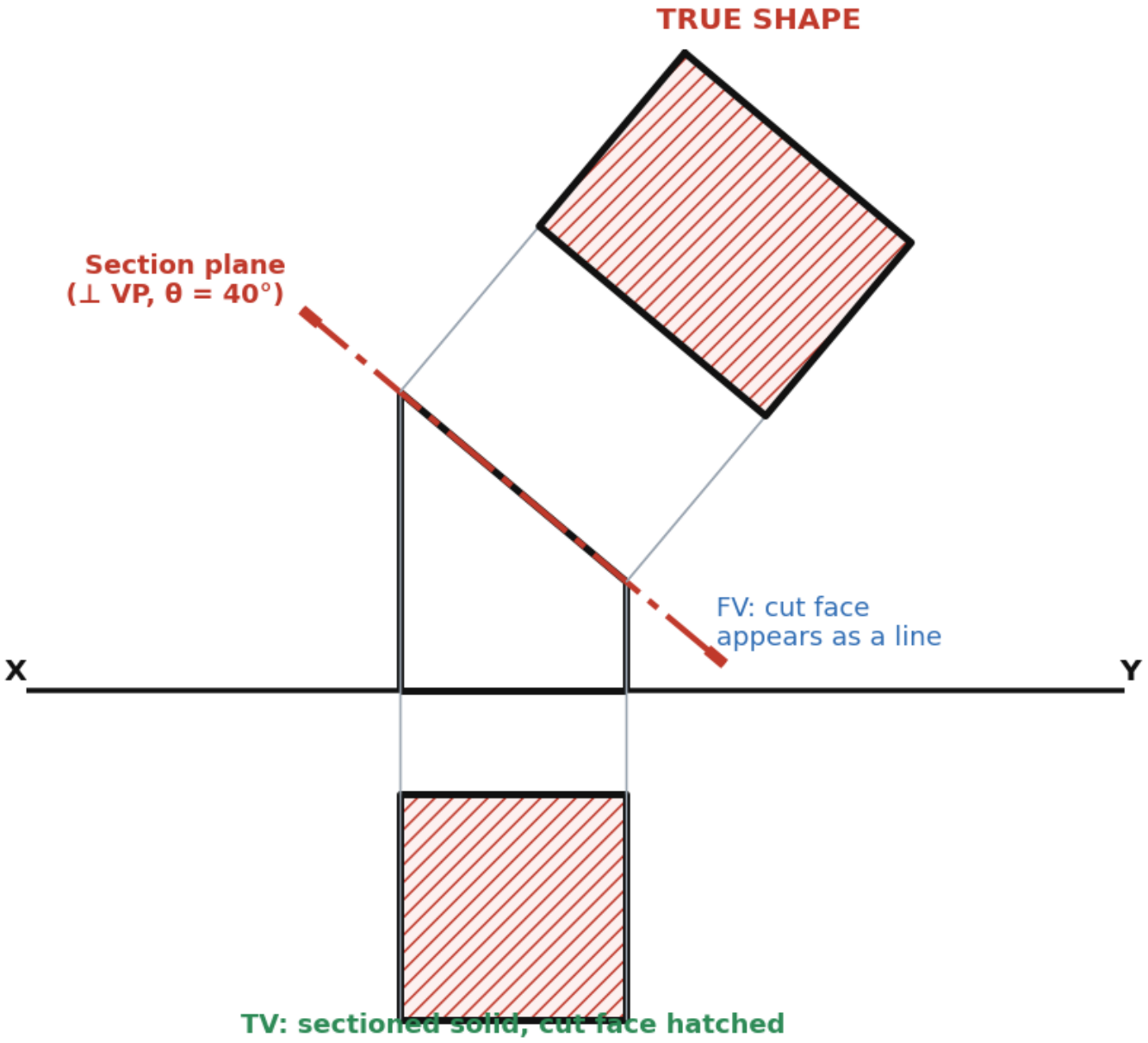


Fig. 2.1 — Square prism cut by an inclined plane (AIP): FV, TV (hatched) and true shape

KEY POINTS

- A plane ⊥ VP and inclined to HP cuts a **prism** in a figure of the **same number of sides as the base** (here a rectangle).
- The TV of the section lies **inside the base outline**; its true length is the **sloping length of the cut line** in the FV (longer than its TV length), while its width equals the base width.
- Distances for the true shape are taken from the **TV** (distance from XY) – the FV only supplies the direction.

Square pyramid (40 × 70) cut by a plane ⊥ VP inclined 35° to HP

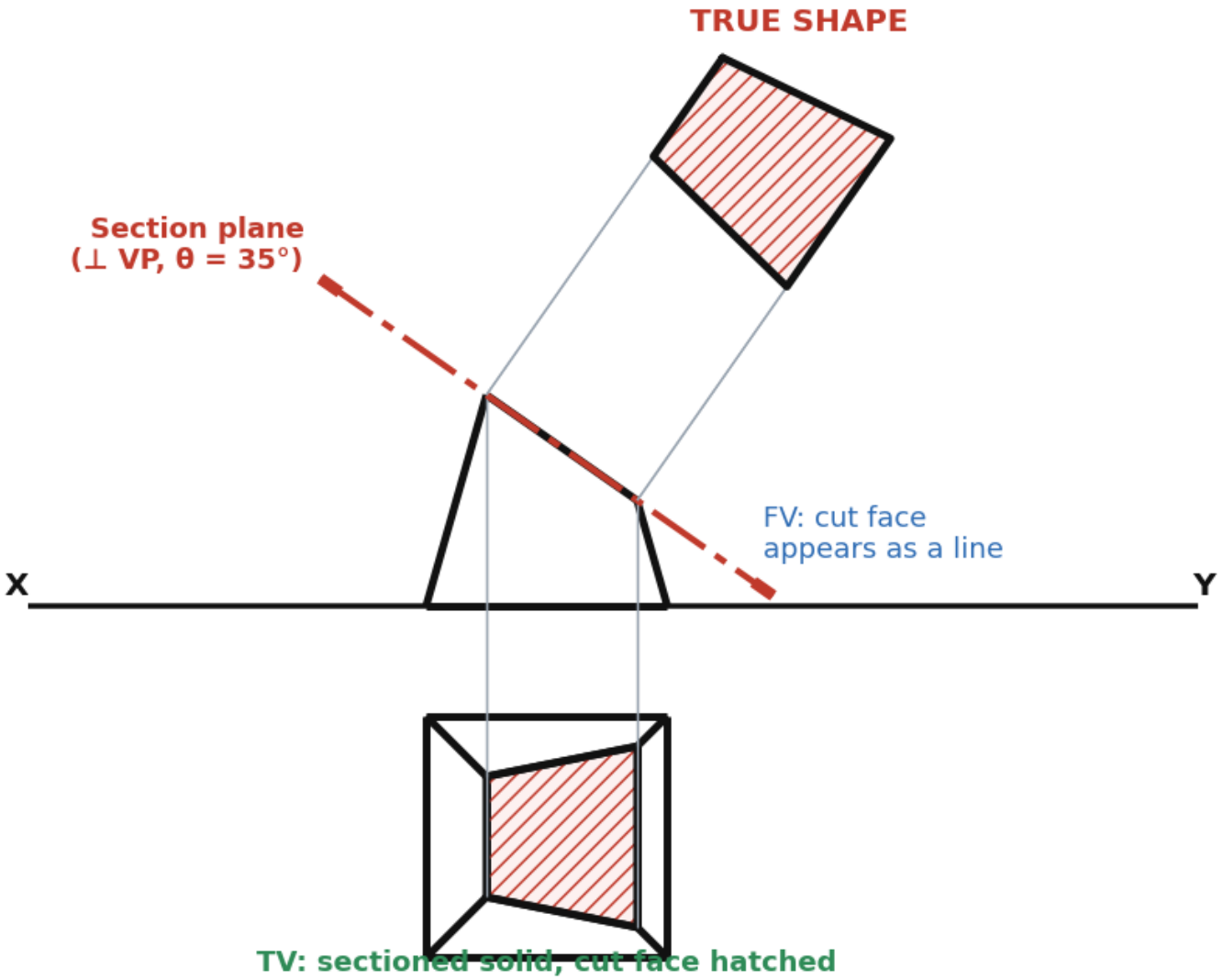


Fig. 2.2 — Square pyramid cut by an inclined plane

For the **pyramid**, each cut point lies on a slant edge; its position in the TV is found by **projecting the FV point to the corresponding edge** (apex-to-corner line). If the plane cuts the base, include the points where it meets the base edges.

Cone (r = 25, h = 65) cut by a plane ⊥ VP inclined 38° to HP

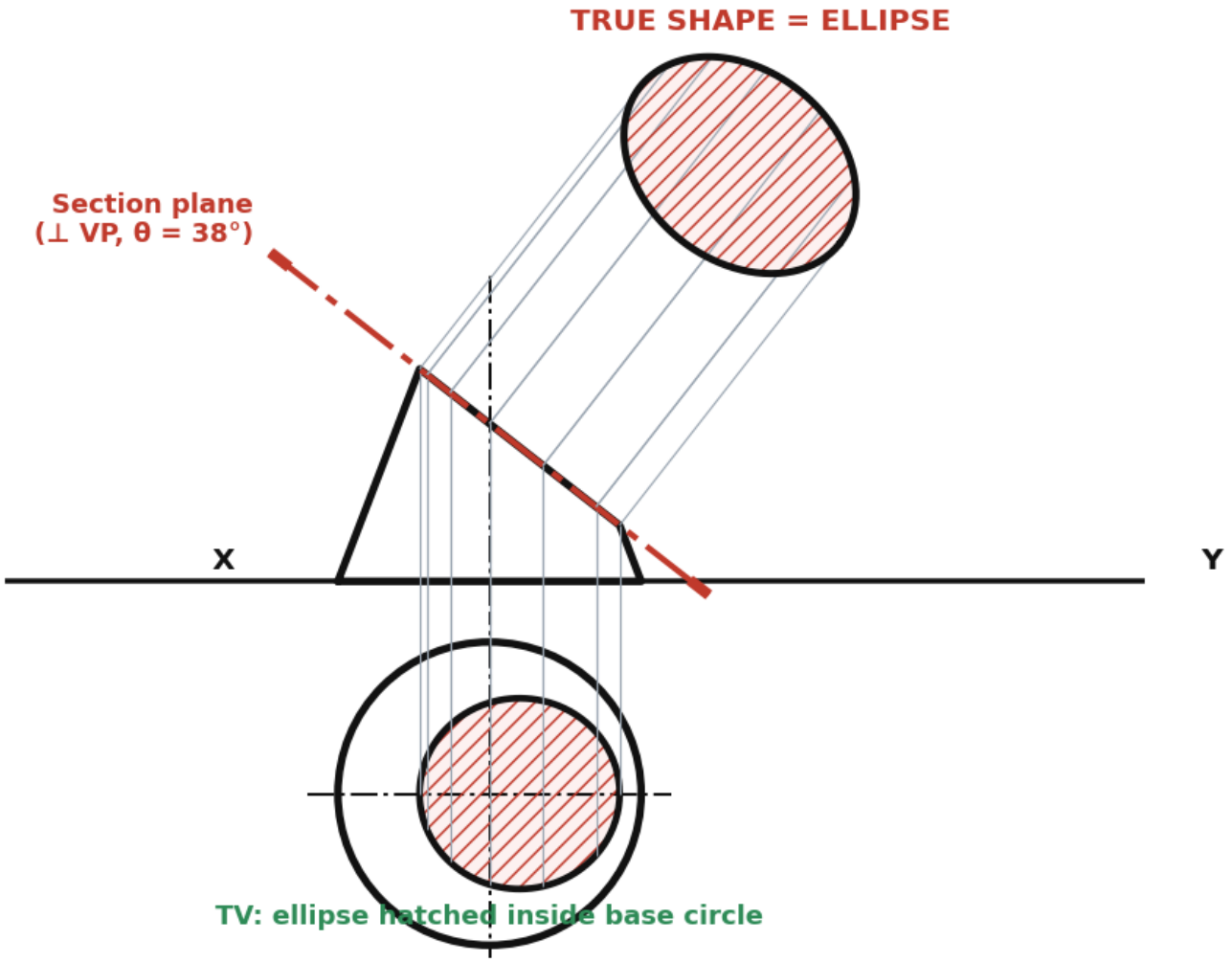


Fig. 2.3 — Cone cut by an inclined plane – the true shape is an ellipse

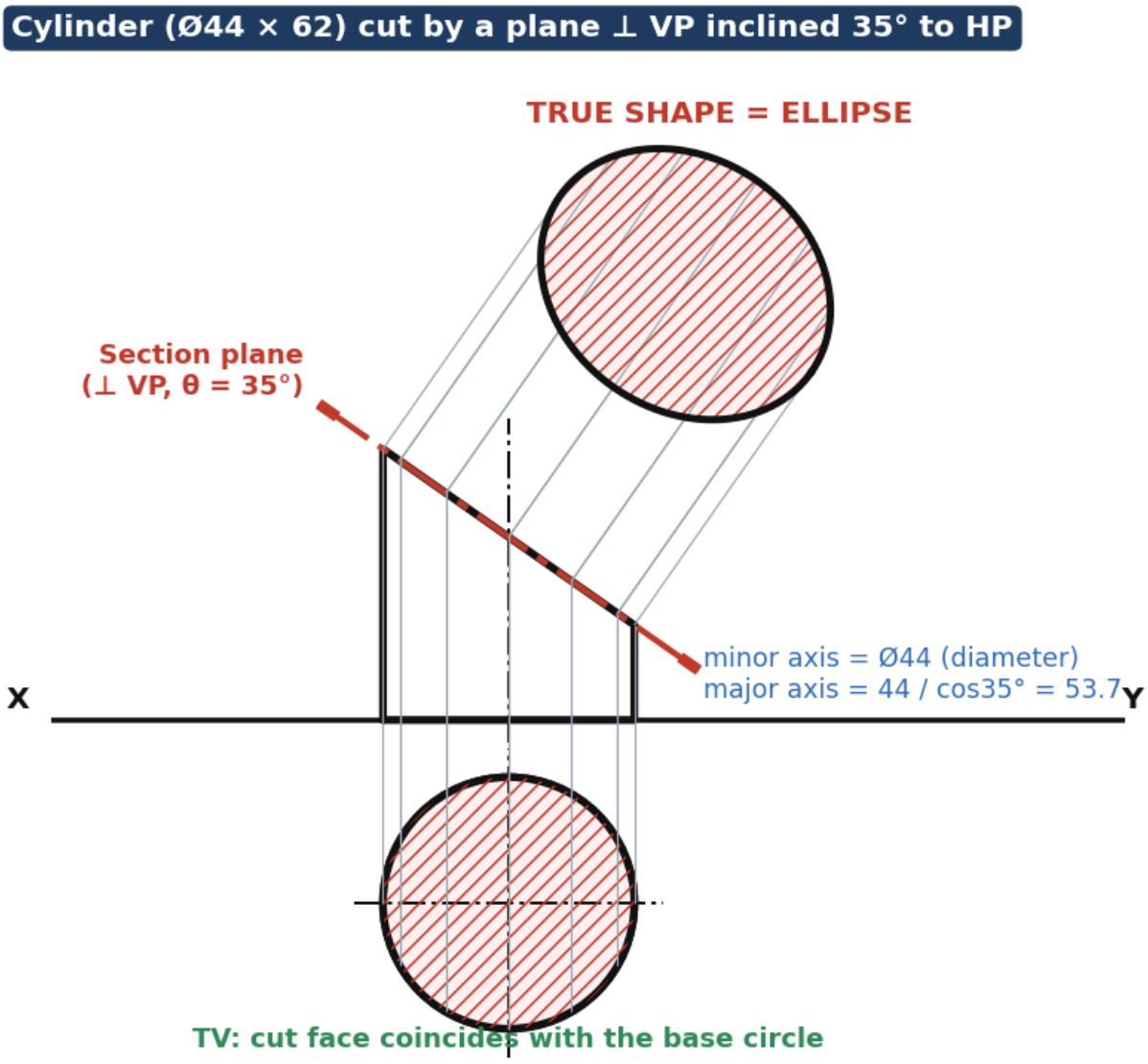


Fig. 2.4 — Cylinder cut by an inclined plane

Cutting plane on a cone	Section	Condition
⊥ axis	Circle	Parallel to base
Inclined, cuts all generators	Ellipse	θ is **less steep** than the generators (θ < angle of generator with base)
∥ one generator	Parabola	θ = angle of generator
Steeper than generators (incl. ∥ axis)	Hyperbola	θ > angle of generator
Through the apex	Triangle	Contains the apex

EXAM TIP

Cylinder: plane inclined to the axis cuts an **ellipse**; plane ∥ axis cuts a **rectangle**; plane ⊥ axis cuts a **circle**. Divide the base circle into 12 equal parts and draw 12 generators – project each cut point to get the ellipse.

SOLVED EXAMPLE

A square pyramid (base 40, axis 70) is cut by a plane ⊥ VP inclined 35° to HP through a point 28 mm up the axis. Find the section.

Cutting line in FV: $z = 28 - (x - x_{axis})\cdot\tan35°$. It meets the left and right slant edges at heights **35 mm** and **17.5 mm**; the length of the cut line in the FV (= true length of the section) is $\sqrt{(25^2 + 17.5^2)} \approx$ **30.5 mm**. Project the points to the TV on the same edges → quadrilateral section; draw the true shape in the auxiliary view, hatch it, and complete the remaining solid.

3. Sectional Views & Hatching

Sectional view: solid cut by plane A–A, the cut metal is hatched

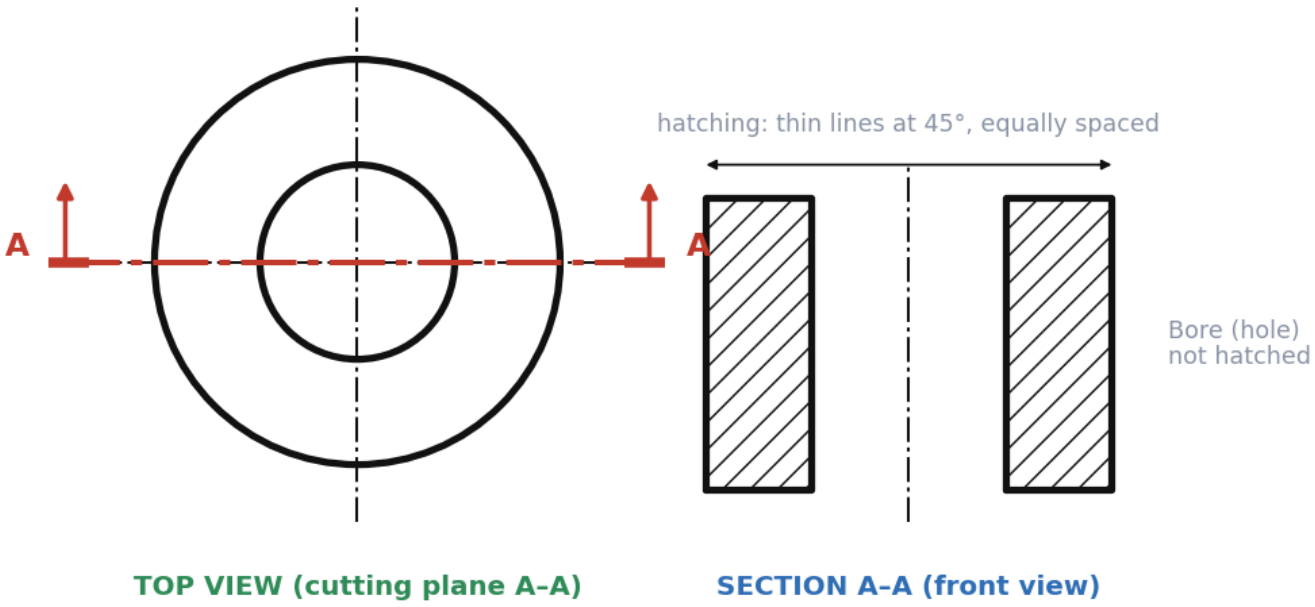


Fig. 3.1 — Sectional view of a bush: cutting plane A–A and hatched section

- **Cutting plane line**: chain thin, thick at ends and where direction changes; arrows show the direction of viewing; label A–A.
- **Hatching**: thin continuous lines at **45°**, equally spaced (1.5 – 3 mm). The same direction & spacing for one part; opposite direction for adjacent parts.
- Holes, voids and the **space** where nothing is cut are **not hatched**; hidden lines are omitted in the sectional view unless needed.
- Ribs, webs, shafts, bolts, keys are **not sectioned** when the cutting plane passes longitudinally through them.

APPLICATIONS

Sectional views reveal **internal features** of engine blocks, valves, pumps, bearings and pressure vessels – without cluttering the drawing with hidden lines.

4. Development of Surfaces

DEFINITION

Development is the unfolding of the lateral (and base) surfaces of a solid on a plane, so that every face appears in its **true shape and size**, laid flat in one piece. It is the reverse of folding a sheet into a solid.

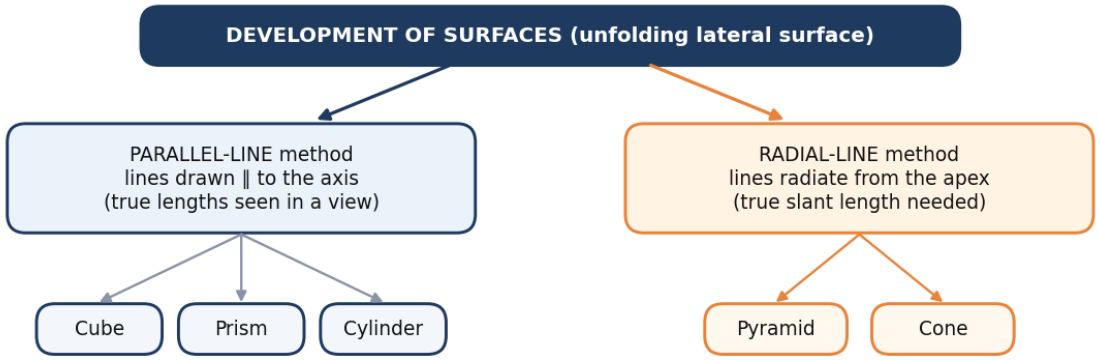


Fig. 4.1 — Methods of development

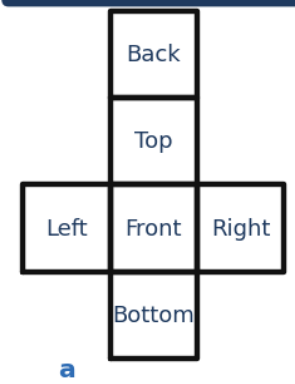
FORMULAS

Lateral surface area (prism) = **perimeter of base × height** | Cylinder: **πDh** | Pyramid: **½ × perimeter × slant height** | Cone: **π r l**

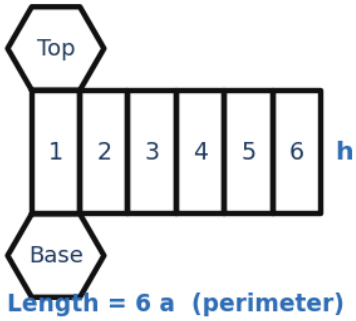
Slant edge of regular pyramid: **l = √(h² + R²)**, R = circumradius of base (square: a/√2) | Cone: **l = √(r² + h²)**, sector angle **θ = (r / l) × 360°**

4.1 Parallel-line development

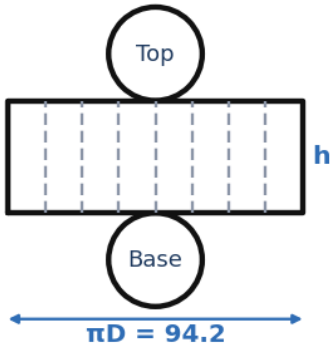
(a) Cube – 6 squares (cross shape)



(b) Hexagonal prism – rectangle 6a × h



(c) Cylinder – rectangle πD × h



(d) Quick reference

Solid	Development	Lateral area
Cube	6 squares	4 a²
Prism	rectangle P × h	P × h
Cylinder	rectangle πD × h	π D h
Pyramid	sectors of triangles	½ × P × l
Cone	sector, angle = (r/l)·360°	π r l

Fig. 4.2 — Development of cube, hexagonal prism and cylinder

PROCEDURE

1. Draw the stretch-out line equal to the **perimeter** (cube 4a, prism n·a, cylinder πD) and mark the divisions of the base edges / 12 equal parts of the circle.
2. Erect **perpendiculars of height h** at each division; join the tops – this is the lateral surface.
3. Attach the **top and base** (true shape) to one edge of the rectangle; for cylinders attach them tangentially at the middle.
4. Mark fold lines as thin dashed lines and show edge numbers.

4.2 Radial-line development

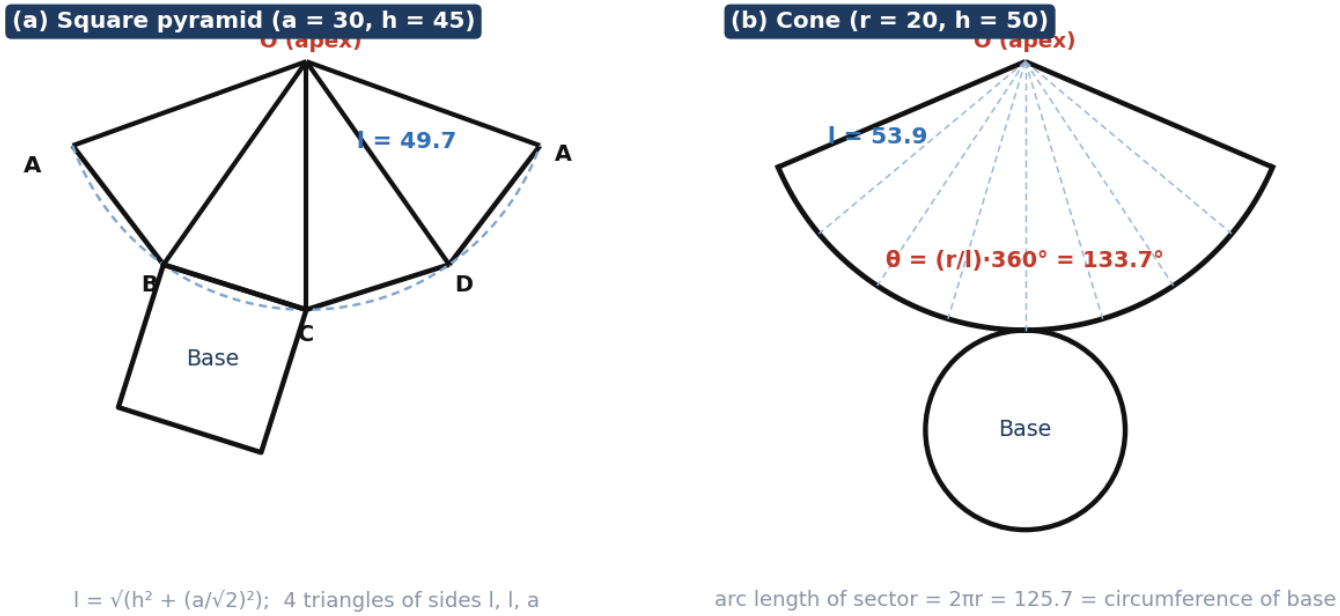


Fig. 4.3 — Development of a square pyramid and a cone

PROCEDURE

1. Find the **true slant length l** (draw the FV of the slant edge ∥ to VP or calculate it).
2. **Pyramid:** with apex O and radius l draw an arc; step off the **base-edge length a** chords **n times**; join to O. Attach the true-shape base to one edge.
3. **Cone:** with radius l draw a sector of angle **θ = (r/l)·360°**; its arc length equals the base circumference 2πr. Attach the base circle tangentially.
4. For a **frustum**, subtract the smaller sector (radius l′ = l × r_2/r_1) from the larger one.

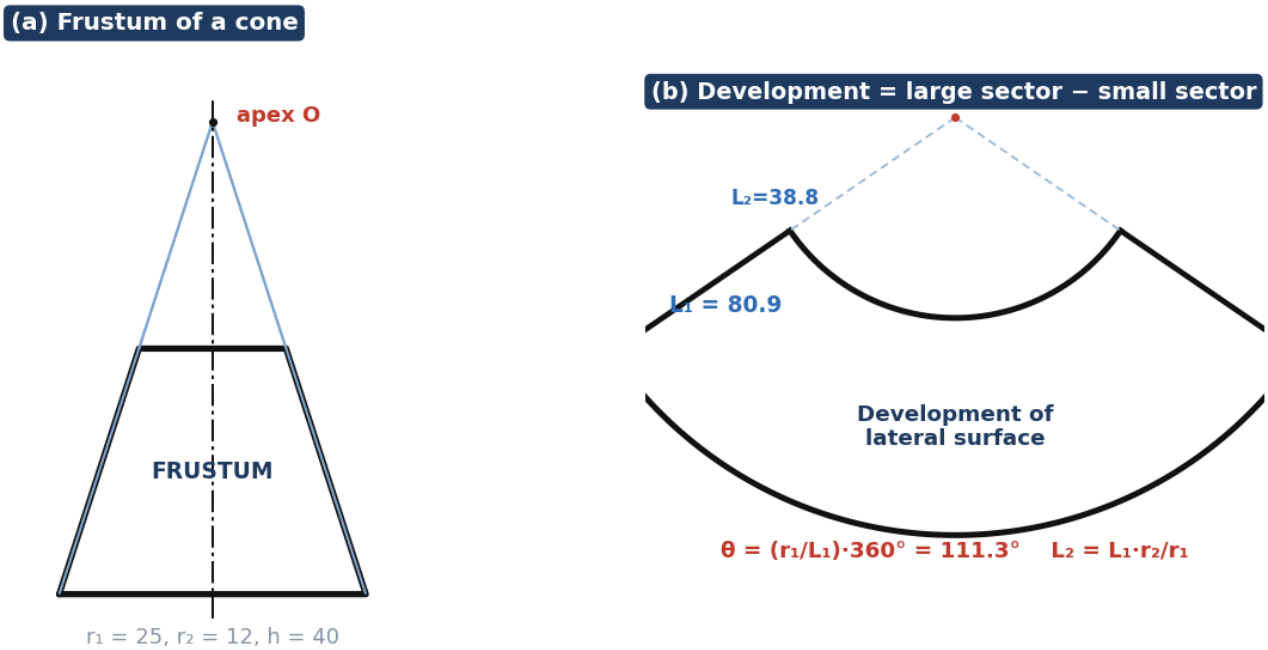


Fig. 4.4 — Development of a frustum (truncated cone)

SOLVED EXAMPLE

Development of a cone, base diameter 40 mm, axis 50 mm.

r = 20; l = √(20² + 50²) = **53.9 mm**; θ = (20 / 53.9) × 360° = **133.6°**. Draw a sector of radius 53.9 mm and angle 133.6°; arc length = 2π × 20 = 125.7 mm ✓.

APPLICATIONS

Development is the basis of **sheet-metal work**: ducts, funnels, hoppers, chimneys, tanks, packaging cartons and pattern-making for fabrication.

5. Quick Revision & Exam Questions

QUICK SUMMARY

- Section ∥ HP → true shape in TV; section ∥ VP → true shape in FV; otherwise use an **auxiliary view**.
- Hatch the cut face at **45°**; cutting plane line = chain thin, thick ends.
- Plane inclined to a cone's axis → **ellipse**; ∥ generator → parabola; steeper → hyperbola.
- Parallel-line: prisms & cylinders (rectangle). Radial-line: pyramids & cones (sectors).
- Cone sector angle θ = (r / l) × 360°; cylinder rectangle = πD × h.

EXAM-ORIENTED QUESTIONS

Q1. Define section plane, sectional view and true shape of a section. **[3]**

Q2. A square prism (side 40, height 70) has its axis ⊥ HP. A plane ⊥ VP and inclined 45° to HP cuts the axis at 35 mm from the top. Draw the FV, sectional TV and true shape. **[8]**

Q3. A hexagonal pyramid (side 30, axis 65) is cut by a plane ⊥ VP inclined 40° to HP bisecting the axis. Draw FV, sectional TV and true shape. **[8]**

Q4. A cone (base Ø50, axis 70) is cut by a plane inclined 45° to HP, passing through the axis midpoint. Draw the sectional views and true shape. **[8]**

Q5. Draw the development of a cube of side 35 mm. **[6]**

Q6. Develop the lateral surface of a hexagonal prism (side 25, height 50). **[6]**

Q7. Develop a cylinder Ø40 × 60; show the development of a point 20 mm from the base on a generator. **[8]**

Q8. Develop a square pyramid, base 40 mm, axis 60 mm. **[8]**

Q9. Develop a cone of base Ø50 and axis 70 mm. Find the sector angle. **[8]**

ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT V

Conversion of Views

Read a drawing like a picture; draw a picture like a drawing.

SYLLABUS COVERED

- ▸ Isometric projection basics
- ▸ Isometric scale, axes & circles
- ▸ Layout of orthographic views
- ▸ Isometric view → orthographic views
- ▸ Orthographic views → isometric view

1. Isometric Projection – Basics

DEFINITION

Isometric projection is a pictorial (single-view) projection in which the three principal axes of the object are equally inclined (**120° apart**) to the plane of projection, so all three axes are equally foreshortened. “Iso” = equal, “metric” = measure.

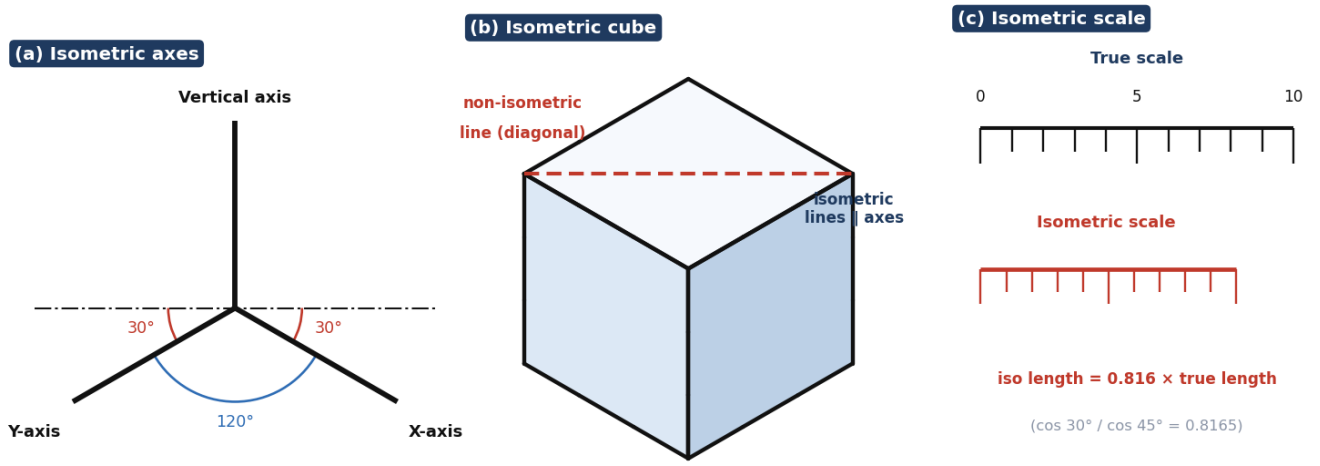


Fig. 1.1 — Isometric axes, isometric cube and isometric scale

Term	Meaning
Isometric axes	Three lines from a point at 120°: one vertical and two at **30°** to the horizontal
Isometric lines	Lines **parallel** to the isometric axes – measured in true (scaled) length
Non-isometric lines	Lines not parallel to the axes (inclined lines) – **not** measured directly; locate their end points
Isometric planes	The three faces of a cube seen in isometric: top, left (front) and right (side)

FORMULAS

Isometric scale = **cos 30° / cos 45° = 0.816** | Isometric length = **0.816 × true length** (isometric *projection*)

Isometric view / drawing uses **full true lengths** on the axes (scale 1) – the picture is 1.225 times larger but identical in shape. Normally used in practice.

EXAM TIP

Remember the three words: **Axes 120°**, **Lines ∥ axes**, **Scale 0.816**. Angles are **not** preserved in isometric – a 90° corner becomes 120° or 60°.

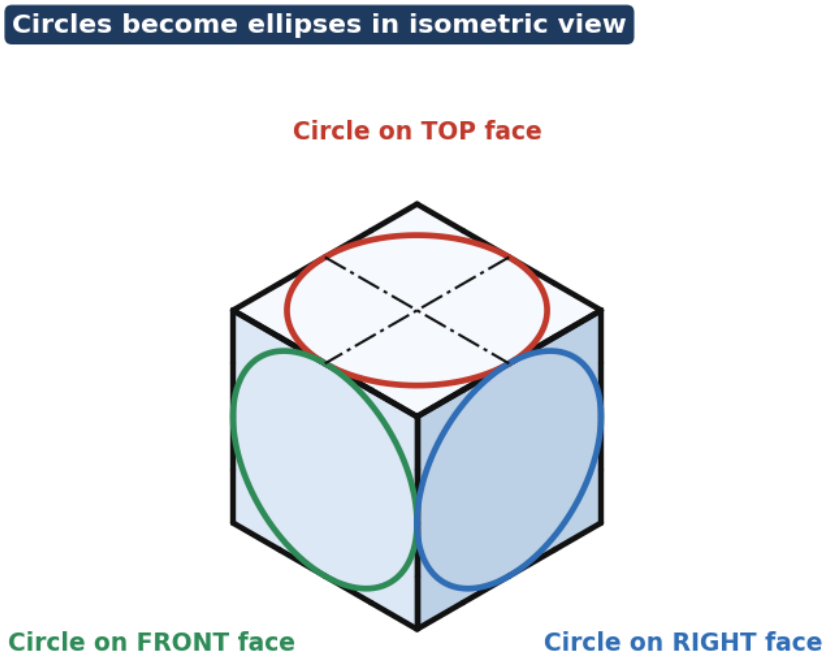


Fig. 1.2 — A circle appears as an ellipse on each isometric plane

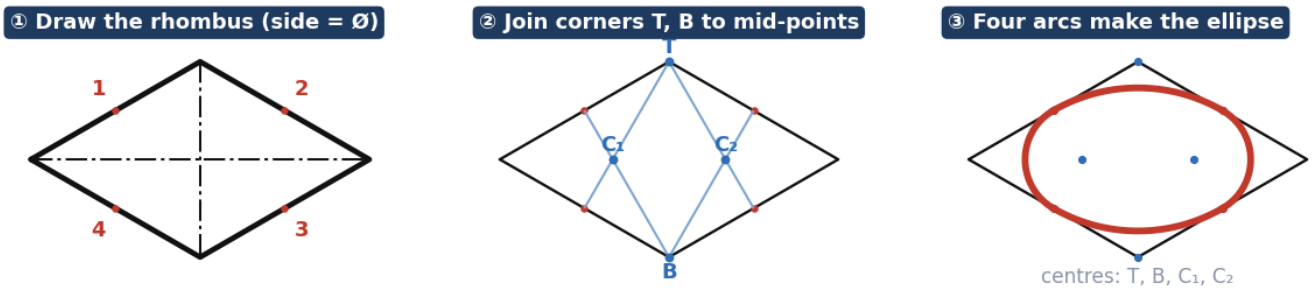


Fig. 1.3 — Four-centre method for an isometric circle

KEY POINTS

- Draw an isometric circle by the **four-centre method** inside the isometric rhombus (side = diameter): join the obtuse-angle corners to the mid-points of opposite sides – their intersections are the centres of the arcs.
- The ellipse's **major axis ⊥ to the axis normal to that plane**: on the top face it is horizontal; on left/right faces it leans along the vertical-ish direction.
- A sphere always appears as a **circle of the true diameter** (use isometric scale: diameter × 1.22 if drawn to isometric view scale).

2. Orthographic Views – Layout & Conventions

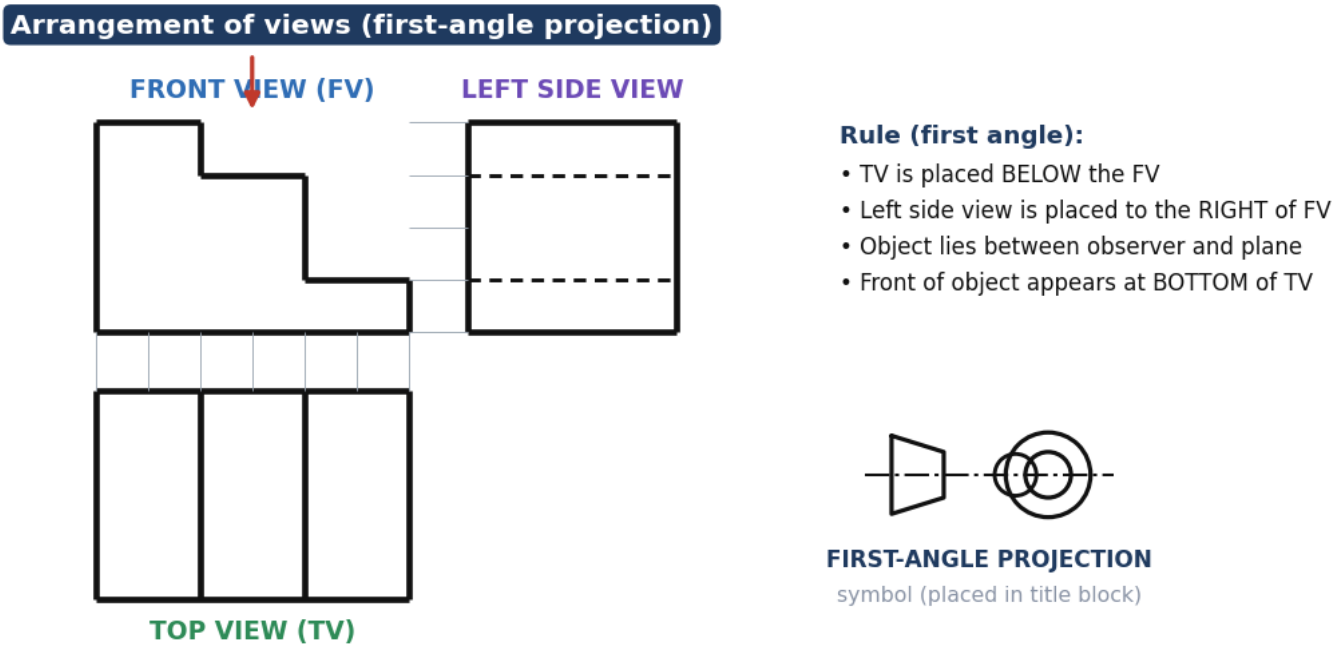


Fig. 2.1 — Arrangement of FV, TV and LSV in first-angle projection

View	Looking from	Placed	Shows
Front view (FV)	Front	Top-left of sheet	Height & width
Top view (TV)	Above	**Below** FV, aligned vertically	Width & depth
Left side view (LSV)	Left side	**Right** of FV, aligned horizontally	Height & depth

Line in a view	What it represents
Thick continuous	An edge or boundary of a **visible** face
Dashed (hidden)	An edge **behind** a face – drawn where the edge is hidden from the viewer
Chain thin	Axis of a hole / cylinder, or symmetry line
A closed rectangle inside a view	A pocket, hole or slot (seen from the side where it opens)
Single line between two faces	A **change of level / surface** (step, slope, or edge between two faces)

EXAM TIP

Projection rules: **(1)** FV & TV have the same **width**, aligned vertically. **(2)** FV & LSV have the same **height**, aligned horizontally. **(3)** TV & LSV have the same **depth** – the front of the object is at the **bottom of TV** and the **right of LSV**.

3. Isometric View → Orthographic Views

Isometric → Orthographic

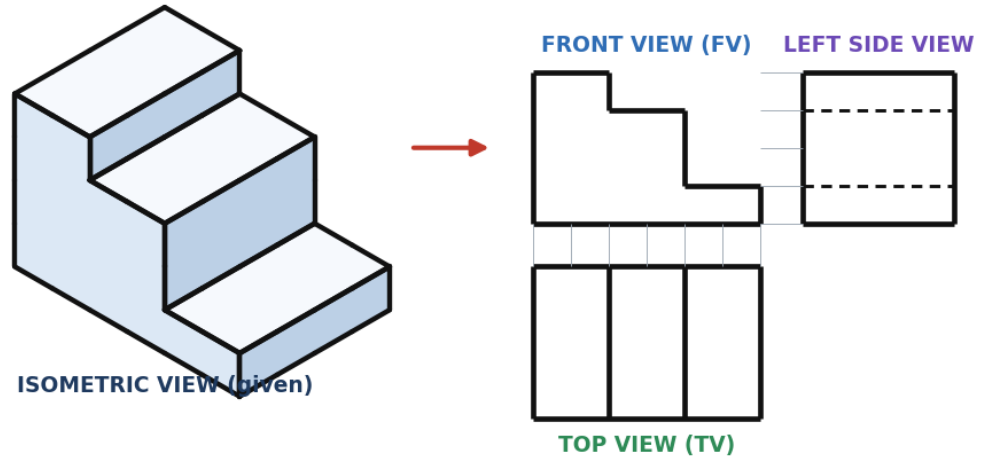
1. Study the isometric view: identify X, Y, Z directions
2. Choose FV direction (the most descriptive face)
3. Draw FV: heights from the vertical edges, widths ∥ X
4. Project TV below FV and LSV to the right of FV
5. Add hidden lines (dashed) & centre lines
6. Dimension & label

Orthographic → Isometric

1. Read the 3 views together: find overall L × B × H
2. Draw the ENCLOSING BOX in isometric (30° lines)
3. Mark features from FV on the front face, from TV on the top face
4. Cut / add portions; draw edges ∥ to iso axes only
5. Circles → ellipses; slant lines → join end points
6. Erase construction lines

Fig. 3.1 — Procedure for conversion in both directions

Example 1 – stepped block



Steps in FV; TV shows step lines; LSV: hidden (dashed) lines of lower steps

Fig. 3.2 — Example 1 – stepped block

PROCEDURE

1. **FV direction:** choose the face that shows the **most shape** (here the left face that shows the steps). Draw FV with correct heights and widths.
2. **TV:** project down from FV; width stays the same, depth = length of the y-axis lines. Each **step** appears as a **line** in TV.
3. **LSV:** project right; **lower steps lie behind** the tallest face → shown by **hidden (dashed) lines**.
4. Check: every point on the isometric must appear in all three views at the aligned positions.

Example 2 – block with rectangular through-slot

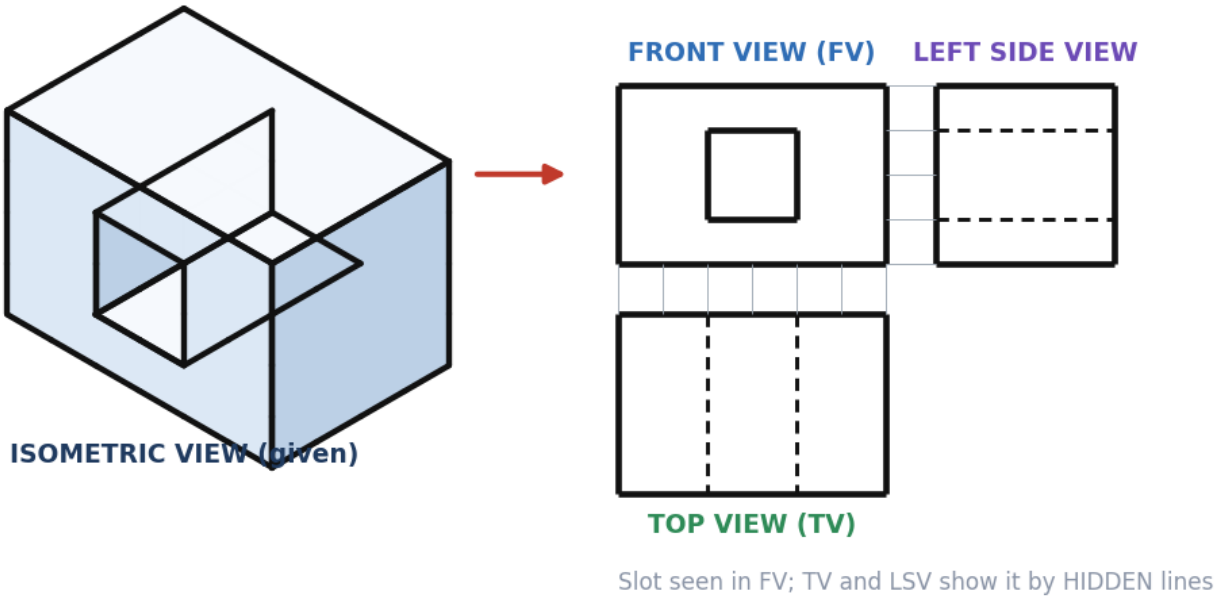


Fig. 3.3 — Example 2 – block with a rectangular through-slot

Example 3 – block with notch & step

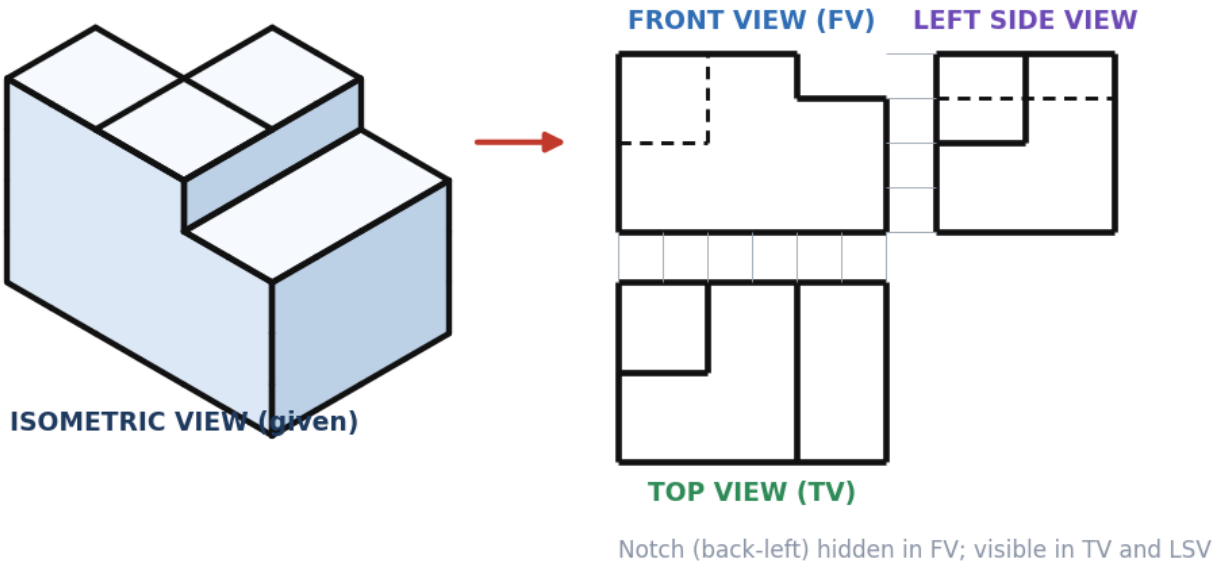


Fig. 3.4 — Example 3 – block with a notch (hidden in one view, visible in others)

Example 5 – inverted-T block with a top slot

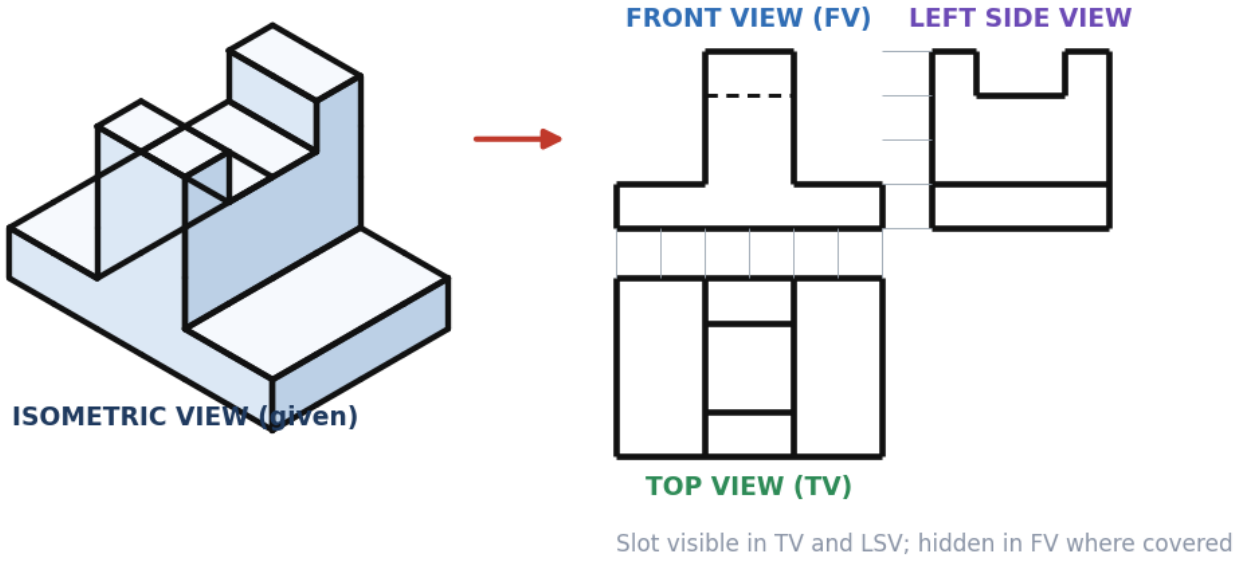


Fig. 3.5 — Example 4 – inverted-T block with a slot in the upright

KEY POINTS

- A feature that is **open to the observer** is drawn with **thick** lines; a feature **inside or behind** is drawn **dashed**.
- A through-slot appears **solid** in the view it passes through and **dashed** in the other two.
- Count the **faces**: the number of regions in each view should match the faces you can see from that direction.

4. Orthographic Views → Isometric View

Orthographic → Isometric (example)

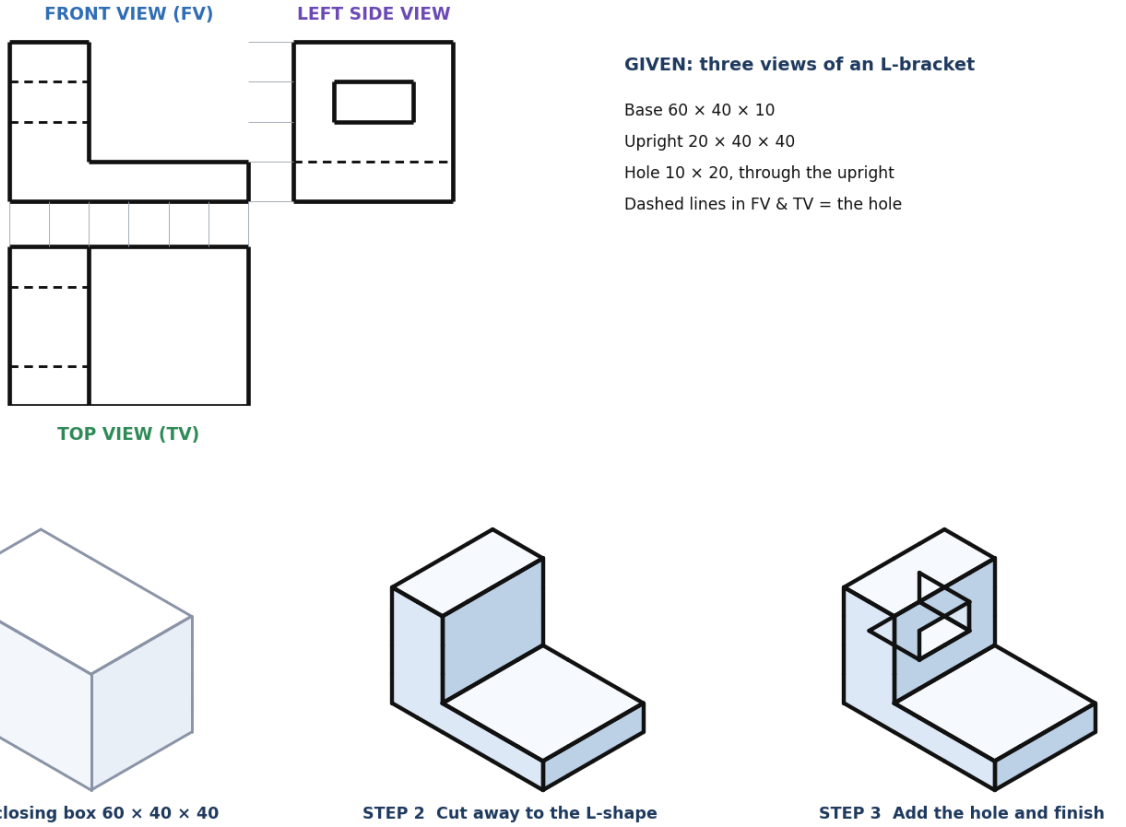


Fig. 4.1 — Example 5 – L-bracket with a through-hole

PROCEDURE

1. Find overall **L, B, H** from the views (FV gives L & H; TV gives L & B).
2. Draw the **enclosing box** using lines at **30°** (X and Y directions) and vertical (Z). Use thin construction lines.
3. Transfer features: **FV → front face** (x–z plane), **TV → top face** (x–y plane), **LSV → side face** (y–z plane). Measure along isometric lines only.
4. Remove material / add parts. For **inclined edges** locate their two end-points on isometric lines, then join them.
5. Make final edges thick; **do not show hidden lines** in an isometric view (unless required); erase construction lines.

Feature in orthographic views	In isometric
Step / shoulder	Offset along the iso lines by step height / depth
Through hole (rectangular)	Draw its opening on the face with iso lines; show the visible inner wall
Circular hole / boss	Isometric ellipse (four-centre) on that face
Inclined (chamfer / slope)	Non-isometric line: join two located end points
Symmetry line	Draw ‖ to iso axis through the centre

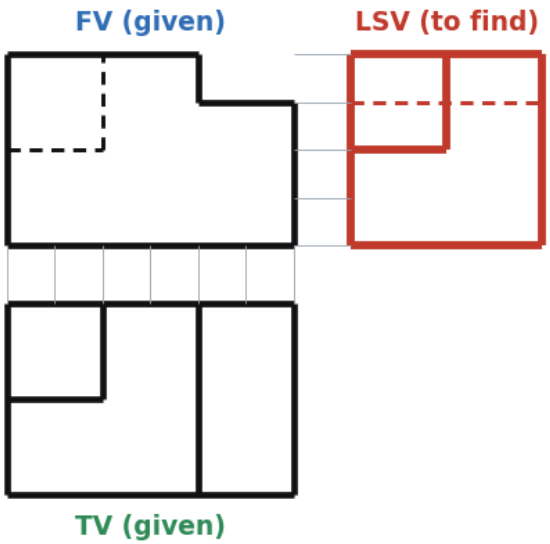
SOLVED EXAMPLE

Quick numerical check. Cube of side 40 mm: isometric *view* edge = 40 mm; isometric *projection* edge = 40 × 0.816 = **32.7 mm**; diagonal of a face (non-isometric line) is not measured – locate its end points.

APPLICATIONS
Conversions are used in **machine-part drawings, CAD model checking, assembly instructions, furniture and architectural plans** – to understand how a part looks and how it is made.

4.1 Missing-view problems

Missing-view problem: draw the LSV from the given FV and TV



Method: take HEIGHTS from the FV, take DEPTHS from the TV (measured from the XY side = back of object), and show the hidden notch with dashes.

Fig. 4.2 — Given two views, find the third

Visualisation tip	How it helps
Number the corners in all views	Tracks every point between FV, TV and LSV
Think of each closed region as a face	A region in one view = a surface; decide whether it is horizontal, vertical or inclined from the other view
A line in one view with a point in another	A face ⊥ to that plane / an edge ⊥ to the plane
Imagine cutting the object with the box lines	Helps to “see” the missing portion
Sketch a rough isometric first	Checks that all three views agree

5. Common Mistakes & Revision

- Using **angles** directly in isometric – always locate end points with lengths along the three axes.
- Placing the TV **above** the FV, or the side view on the wrong side (this is third-angle).
- Forgetting **hidden lines** or drawing them where the feature is visible.
- Confusing **isometric scale** (0.816) with the **isometric axes angle** (30° / 120°).
- Drawing circles as true circles in isometric – they must be **ellipses**.

QUICK SUMMARY

- Isometric: axes 120°, lines ∥ axes true, scale 0.816, circles → ellipses, no hidden lines.
- First angle: TV below FV, LSV on the right of FV; front of object = bottom of TV.
- Iso → ortho: choose FV, project TV & LSV, add dashed lines.
- Ortho → iso: overall box → features from each view → remove / add → thick final lines.

EXAM-ORIENTED QUESTIONS

Q1. Define isometric axes, isometric lines and isometric scale. State the value of the isometric scale. **[4]**

Q2. Explain the difference between isometric view and isometric projection. **[4]**

Q3. Draw the FV, TV and LSV of the object whose isometric view is shown (stepped block with a slot). **[8]**

Q4. Draw the isometric view of an object whose three orthographic views are given (L-bracket / T-block). **[8]**

Q5. Draw an isometric view of a cylinder Ø40 × 60 standing on its base; show the isometric circles on top. **[6]**

Q6. List the rules for drawing hidden lines in orthographic views. **[6]**

Q7. Draw the views of a block 60 × 40 × 40 with a 20 × 20 notch at one top corner. **[8]**